AutoProject+: A WEB-BASED ONLINE SERVICE BOOKING AND COST ESTIMATION SYSTEM WITH HYBRID CUSTOMER SUPPORT FOR AUTOPROJECT-D CUSTOM GARAGE

A Capstone Project

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CHAPTER 1

The Problem and Its Background

This chapter discusses the background and context of the study and the technologies related to the automotive service industry. It also presents the objectives of the study, scope and limitations, and the significance of the system, providing an overview of the proposed project.

Overview of the Current State of the Technology

In recent years, the evolution of Information and Communication Technologies (ICT) has fundamentally reshaped the operational landscape of service-oriented businesses, prompting a shift from traditional manual processes toward integrated digital frameworks. Agustian et al. (2023) emphasize that digital transformation is no longer optional but has become a strategic imperative for organizations seeking to maintain market relevance and sustain competitive advantage. In the context of service management, web-based systems play a significant role in improving operational efficiency by automating procedures and streamlining administrative tasks. Peras et al. (2024) explain that web-based systems enhance operational efficiency through the automation of administrative procedures, while Aziz et al. (2024) similarly note that digital platforms help reduce manual record-keeping errors and streamline service management processes.

The integration of automated technologies has further expanded the capabilities of modern service management systems. Features such as SMS notifications and online payment portals enable businesses to optimize internal resource utilization while providing customers with transparent and convenient service interactions. These digital tools support the growing expectation of customers for real-time updates and accessible transaction processes (eBillH20, 2025). This shift is further reinforced by Muhammad and Stukalina

(2024), who highlight that modern consumers increasingly expect immediate, data-driven responses throughout their service journey. Such expectations emphasize the importance of adopting digital platforms that support efficient communication and effective service delivery.

Despite these technological advancements, the automotive repair industry, particularly small and medium-sized service shops, continues to rely heavily on manual processes to manage operations and customer interactions. Orderry (2025) notes that independent automotive garages are under increasing pressure to modernize their operations as vehicle technologies continue to evolve. Similarly, Alev Digital (2025) reports that many service shops still rely on traditional booking practices such as phone calls and walk-in transactions despite the growing demand for digital solutions. While social media platforms such as Facebook and WhatsApp provide convenient communication channels, these tools are often fragmented and lack the structured functionality required for professional service documentation and systematic customer data management.

As a result, service providers frequently encounter operational inefficiencies. Scheduling errors, including overlapping appointments and unexpected no-show gaps, can significantly disrupt daily operations and lead to revenue losses for small automotive businesses (Garage360, 2025). Additionally, the absence of transparent digital communication and inconsistent pricing information has been identified as a major factor contributing to customer dissatisfaction and declining service loyalty in recent years. These operational challenges highlight the need for more structured and reliable service management solutions within the automotive repair sector (Cox Automotive, 2025).

In this context, digital transformation offers a promising approach for addressing these operational limitations. Vial (2021) explains that digital transformation enables organizations to improve operational performance and strengthen customer engagement through the integration of modern technologies into business processes. However, within the automotive customization sector, the adoption of dedicated web-based management systems remains relatively limited. Many garages continue to rely on walk-in transactions and social media messaging to manage appointments and service inquiries. Although these methods are familiar and widely used, they do not provide a systematic approach for organizing service schedules, documenting repair details, or securely storing customer information. These capabilities are essential for improving operational efficiency and delivering consistent service quality. Therefore, the development of a web-based automotive service management system may provide a practical solution for enhancing operational efficiency, improving service coordination, and strengthening customer engagement within small and medium-sized automotive repair businesses.

Project Context

AutoProject-D Custom Garage is a well-known automotive customization shop located at Daliz Street, Sariaya, Quezon. The business has been operating for approximately 19 years and is owned and managed by Mr. Makol Pasumbal, who oversees daily operations, manages customer transactions, and supervises all customization and repair services. The garage originated from a professional connection to Project-D, a renowned Japanese automotive customization group. Through this partnership, Mr. Pasumbal was granted exclusive permission to establish the brand in the Philippines, making AutoProject-D the only recognized and authentic branch of Project-D in the country. This unique distinction has contributed significantly to its credibility, customer trust, and established reputation within the local automotive industry.

According to an interview with Mr. Pasumbal, the garage operates with a skilled team consisting of two (2) office staff members and four (4) specialized mechanics who work daily from 8:00 AM to 6:00 PM. Each mechanic possesses a specific area of expertise, including body works, paint application, engine repair, electrical wiring, and general technical support. Despite the limited manpower, the garage manages a high volume of complex projects, which makes systematic coordination and scheduling very important.

AutoProject-D Custom Garage provides a variety of services, ranging from technical performance upgrades like exhaust and intake piping to structural and aesthetic repairs. These services are priced as a baseline for the system's cost engine, with performance tasks such as intercooler installation costing between ₱12,000 and ₱15,000. Aesthetic work, including body repairs and custom paint jobs, ranges from ₱3,000 to ₱20,000 depending on the project's size. For customers wanting a total transformation, the garage offers a premium Full Custom Build Package for ₱35,000 to ₱45,000. By putting these specific prices into the AutoProject+ platform, the garage can replace confusing verbal quotes with clear, data-driven estimates.

Currently, the garage operates on manual processes, where customers book services through walk-ins or social media like Facebook Messenger. Appointments are finalized through informal talks or handwritten notes, which often leads to overlapping schedules and forgotten details. Costs are calculated by hand, often causing pricing mistakes and misunderstandings. Mechanics receive their tasks verbally from the owner, and there are no digital records for past work or money transactions. Without a digital system to track ongoing projects, the shop often takes on more work than the staff can handle, causing delays and workspace congestion.

To address these challenges, the researchers proposed AutoProject+, a web-based online booking and cost estimation system with hybrid customer support designed specifically for AutoProject-D Custom Garage. The system centralizes booking requests into a single platform, provides a more consistent approach to service cost estimation, creates digital job orders, and improves communication between customers and garage personnel through organized and structured processes. The system requires a reservation fee paid through GCash or Maya Qr to confirm a slot. This move from manual process to a structured digital platform allows the garage to keep service quality high while keeping professional records and organized workflows.

This project also helps meet the United Nations Sustainable Development Goal (UNSDG) 8, which focuses on Decent Work and Economic Growth. By using technology to organize job assignments and reduce stress for the mechanics, AutoProject+ makes the business more productive and improves the quality of the work environment. It serves as a real-world example of how local technological improvements can lead to better, more efficient labor standards in the automotive industry.

Objectives of the Project

General Objectives

The general objective of this project is to design and develop a web-based online service booking and cost estimation system with hybrid customer support that improves operational performance and customer engagement for AutoProject-D Custom Garage.

Specific Objectives

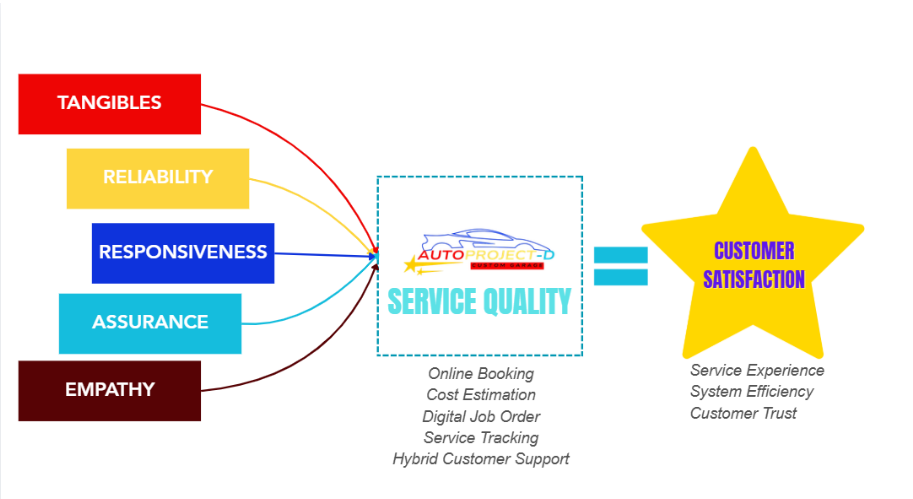
The goal of this research project is:

1. to design a web-based booking and service management system that supports online reservations and walk-in customers through digital scheduling, while implementing role-based access control and customer engagement features such as ticket-based support and service tracking;
2. to develop an automated cost estimation system that generates preliminary service quotations based on vehicle and service details; and
3. to create an administrative monitoring and reporting module that allows administrators to track mechanic activities, monitor service progress, manage customer and service records securely, evaluate labor and sales transactions, and generate booking and revenue reports.

Theoretical Framework

Figure 1

SERVQUAL model by Parasuraman, Zeithaml, and Berry (1988).



The proposed research is grounded in the SERVQUAL Model, a multi-dimensional framework developed by A. Parasuraman, Valarie A. Zeithaml, and Leonard L. Berry (1988) to measure the gap between customer expectations and perceptions of service quality. This model provides a scientific foundation for the study by identifying five key dimensions: reliability, responsiveness, assurance, empathy, and tangibles. In the context of AutoProject-D Custom Garage, the SERVQUAL Model serves as a design blueprint, ensuring that the transition from manual to digital processes directly improves the perceived quality of highly specialized automotive customization services.

The dimension of Reliability is directly applied to the development of the centralized booking and digital scheduling system. By replacing informal handwritten notes with a synchronized digital calendar, the system ensures that service promises are kept and scheduling overlaps are eliminated. This technical reliability is further reinforced by the Mechanic Module, which uses a status-driven queue to ensure that complex job orders are processed systematically without being overlooked.

Responsiveness is integrated into the system through the online reservation functionality and the ticket-based hybrid customer support feature. These technical components allow the garage to provide timely assistance and real-time service tracking updates. By reducing the time customers spend waiting for verbal updates via social media, the system professionally manages the "responsiveness gap" inherent in the current manual communication process.

The development of the automated cost estimation module specifically addresses the Assurance dimension. By generating preliminary digital quotations based on vehicle specifications and service details, the system provides a transparent and standardized pricing framework. This scientific approach to cost estimation builds customer trust and confidence, as users are assured that pricing is based on consistent data rather than subjective spot-calculations.

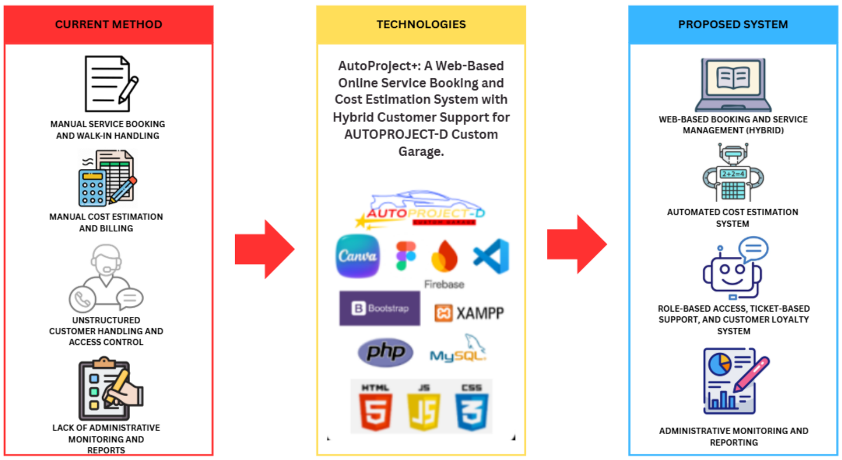
Tangibles in this study are represented by the administrative reporting module and the web-based user interface. While automotive customization is a physical service, the digital records, professional service documentation, and clean administrative dashboards serve as "digital tangibles" that signify professionalism. These structured outputs provide the garage with a modern image of organizational efficiency that complements the physical quality of their customization work.

Finally, the Empathy dimension is reflected in the system's centralized database, which maintains comprehensive customer and vehicle histories. This allows garage personnel to recognize returning clients and understand their specific customization preferences, such as previous paint codes or specific fabrication requirements. By linking these five SERVQUAL dimensions to the core modules of AutoProject+, the theoretical framework ensures that the system development is scientifically designed to enhance both operational efficiency and total customer satisfaction.

Conceptual Framework

Figure 2

Conceptual Framework for AutoProject+



The conceptual framework of the proposed AutoProject+, illustrates the transformation of the existing operational practices of AutoProject-D Custom Garage into a structured, web-based service management system. The framework is divided into three major components: the Current Method, the Technologies Used, and the Proposed System, showing the logical flow from existing challenges to the intended solution.

The Current Method reflects the present operational setup of AutoProject-D Custom Garage, which relies heavily on manual service booking and walk-in handling. Appointment scheduling is conducted through personal visits and informal online messaging, resulting in unorganized schedules and possible booking conflicts. Cost estimation and billing are performed manually through verbal discussions and handwritten computations, which may lead to inconsistent pricing and misunderstandings. Customer handling and access control are unstructured, with no formal role-based authorization or systematic tracking of inquiries and concerns. In addition, the garage lacks administrative monitoring and reporting tools, as service records, sales transactions, and customer activities are not summarized through automated reports. These conditions contribute to inefficient workflow management, limited operational visibility, and reduced service transparency.

The Technologies component represents the tools and platforms used to develop the proposed solution. AutoProject+ is designed as a web-based system utilizing PHP and MySQL for server-side processing and database management, while HTML, CSS, and JavaScript are used to structure and enhance the user interface. Bootstrap is employed to ensure responsive and user-friendly design across devices, and XAMPP serves as the local development and testing environment. Canva is used for interface prototyping and visual layout preparation. These technologies collectively support secure access, organized data processing, and reliable system performance.

The Proposed System illustrates the intended improvements achieved through AutoProject+. The system introduces a Web-Based Booking and Service Management System with Hybrid Support, allowing both online reservations and walk-in customers to be accommodated through digital scheduling and job order management. An Automated Cost Estimation System is implemented to generate standardized service quotations based on vehicle details and selected services, subject to approval prior to execution. The system also includes Role-Based Access, Ticket-Based Customer Support, and a Customer Loyalty System, ensuring structured customer handling, controlled access, and improved engagement. Lastly, Administrative Monitoring and Reporting features provide automated summaries of bookings, service transactions, and operational performance to support informed decision-making.

Scope and Limitation of the Project

The proposed study focuses on the design and development of AutoProject+: A Web-Based Online Service Booking and Cost Estimation System with Hybrid Customer Support for AutoProject-D Custom Garage. The system is intended to serve as a centralized digital platform that improves booking management, cost estimation, service tracking, and internal communication within the garage environment. AutoProject+ will operate through standard web browsers and will support both online reservations and walk-in customers through a structured role-based interface. The system scope is limited strictly to automotive customization services, including custom body works, intake and exhaust piping, and specialized paint jobs, while excluding routine maintenance and emergency roadside assistance.

Under the Customer Module, registered users can create secure accounts to browse customization services, including a specialized paint repair and custom paint job category. To facilitate accurate remote assessments, the module includes an image upload feature where customers can submit photos of vehicle damage for administrative evaluation. Customers will input specific vehicle models and parts requirements to receive preliminary cost estimates, which serve as reference values until a physical inspection is conducted. The system promotes transparency and trust by allowing customers to acknowledge approved estimates digitally before work begins. To reduce paperwork and improve communication, the system implements automated SMS notifications triggered by key activities such as booking approvals, payment confirmations, and service completion. The hybrid customer support feature combines automated status tracking with a manual ticket-based inquiry system where staff can provide personalized responses during business hours.

The system incorporates a secure payment and reservation functionality to formalize appointments. Customers are required to pay a reservation fee via integrated GCash and Maya e-wallet services. To ensure financial accountability, the system reflects payment confirmation only after the administrator validates the transaction against the garage e-wallet records. These fees are non-refundable if the customer fails to appear on the confirmed date without prior coordination. For walk-in clients, authorized staff members will encode all vehicle and service details into the platform. An inventory management feature is included to help staff identify the availability of specific customization parts and inform clients of necessary lead times if parts are currently out of stock.

Under the Mechanic Module, authorized personnel log in to access a status-driven job queue that organizes assignments as queued, ongoing, paused, or completed. This module optimizes mechanic workload and prevents overbooking by providing a real-time visual buffer of ongoing tasks, allowing the administrator to distribute assignments based on specific mechanic expertise in bodywork or electrical wiring. The Administrator Module, managed by the garage owner Mr. Makol Pasumbal, retains full control over user roles, booking approvals, and the finalization of service costs. This module generates a comprehensive list of analytical reports, including booking summaries, service status reports, cost estimation histories, and revenue performance reports, which assist in data-driven decision-making and operational monitoring.

Despite its operational benefits, the system has defined limitations. AutoProject+ does not guarantee automatic booking approvals, exact completion timelines, or real-time mechanic GPS tracking. The cost estimation module provides a preliminary quote that is subject to change following a mandatory physical vehicle inspection prior to admitting the car for repair. The platform does not include integration with external accounting systems, tax computation modules, or payroll management. The system relies on manual inputs for inventory updates and service progress to maintain accountability. Furthermore, the system requires stable internet connectivity, and any delays caused by external factors such as parts availability or manpower limitations remain beyond the control of the digital platform.

Significance of the Project

The proposed project, AutoProject+: A Web-Based Online Service Booking and Cost Estimation System with Hybrid Customer Support for AutoProject-D Custom Garage, is significant as it seeks to address the operational challenges encountered by small to medium-sized automotive customization garages that rely on manual and semi-digital processes. By proposing a structured and centralized web-based platform, the study aims to improve service efficiency, transparency, and customer engagement.

For AutoProject-D Custom Garage, the proposed system is expected to provide a centralized platform for managing service bookings, cost estimation, job orders, and customer communication. Once implemented, AutoProject+ may help reduce scheduling conflicts, minimize idle service slots caused by no-show customers, and improve workflow organization. This initiative directly supports the United Nations Sustainable Development Goal (UNSDG) 8: Decent Work and Economic Growth by utilizing technology to upgrade operational productivity. The proposed use of digital job orders and service tracking is intended to support better operational control and more organized garage management.

For Customers, the proposed system aims to provide a more convenient and transparent service experience by allowing them to submit booking requests, receive preliminary cost estimates, and monitor service status through a web-based platform. The integration of electronic payment options for reservation fees ensures a formalized commitment, while the inclusion of ticket-based customer support and service documentation is expected to reduce miscommunication and improve customer trust and satisfaction.

For Administrators, Staff, and Mechanics, AutoProject+ is proposed to assist garage personnel by simplifying booking approvals, job assignment, service monitoring, and customer record management. The implementation of role-based access is intended to promote accountability and improve job quality by providing mechanics with clear, documented tasks rather than verbal instructions. By reducing manual errors and improving coordination among personnel, the system fosters a more professional and less stressful work environment.

For the Researchers, this proposed project will allow the researchers to apply theoretical knowledge in web development, database management, system analysis, and software engineering to a real-world scenario. The study is expected to enhance the researchers' technical, analytical, and problem-solving skills through system design and development planning within the framework of the Agile Scrum Methodology methodology.

For Future Researchers, the proposed system may serve as a reference for future studies related to web-based service management systems, booking platforms, and digital transformation in small-scale enterprises. Future researchers may build upon this study by extending its features or adapting the proposed framework to other service-oriented industries that require automated estimation and structured scheduling.

Definition of Terms

The following are operationally and technically defined to facilitate a more comprehensive understanding of this study.

Administrator refers to the proposed system user role responsible for managing service bookings, reviewing cost estimates, configuring system settings, and overseeing overall system operations once the system is implemented.

Authentication refers to the proposed security process used to verify the identity of users before granting access to the system.

AutoProject+ refers to the proposed web-based online service booking and cost estimation system with hybrid customer support developed for AutoProject-D Custom Garage.

AutoProject-D Custom Garage refers to a local automotive customization garage located in Sariaya, Quezon, which serves as the proposed client of the study.

Booking Request refers to a service request submitted by customers through the proposed system, containing vehicle details, selected services, and preferred schedules, subject to approval by authorized personnel.

Bootstrap refers to a front-end framework proposed for designing responsive and mobile-friendly web interfaces for the system.

Canva refers to a design tool proposed for creating interface prototypes, layouts, and visual materials used in system development.

Cost Estimation refers to a proposed system feature that generates preliminary service quotations based on selected services, vehicle type, and preferred parts.

Customer refers to a vehicle owner who uses the proposed system to request automotive services, review cost estimates, and monitor service progress.

CSS (Cascading Style Sheets) refers to a styling language proposed for defining the visual layout, appearance, and design of the web-based system.

Database refers to a structured data repository proposed for storing user information, booking records, service details, and transaction data of the system.

Digital Job Order refers to a proposed electronic service record used to document approved services, assigned mechanics, and the current service status.

HTML (Hypertext Markup Language) refers to the standard markup language proposed for structuring and presenting web content within the system.

Hybrid Customer Support refers to a proposed customer assistance approach that combines ticket-based online communication with in-person service coordination.

JavaScript refers to a programming language proposed for implementing interactive features and dynamic behaviors within the web-based system.

Mechanic refers to a proposed system user responsible for performing assigned automotive services and updating job progress within the system once implemented.

MySQL refers to a relational database management system proposed for storing and managing structured system data.

PHP (Hypertext Preprocessor) refers to a server-side scripting language proposed for handling system logic, data processing, and database interaction.

Role-Based Access Control refers to a proposed security mechanism that restricts system access and functionalities based on assigned user roles.

Service Status refers to the stage of a service request, such as queued, ongoing, or completed, which is displayed by the proposed system.

Ticket-Based Communication refers to a proposed system feature that allows customers to submit inquiries or concerns that can be monitored, tracked, and resolved by authorized personnel.

Trust Score refers to a proposed system-generated indicator that reflects customer booking reliability based on attendance records and service history.

Walk-In Customer refers to a customer who visits the garage without prior online booking and whose service details are recorded in the system by authorized personnel.

Web-Based Application refers to a software system designed to be accessed through a web browser without requiring local installation on the user's device.

XAMPP refers to a local server environment consisting of Apache, MySQL, PHP, and Perl, commonly used for system development and testing.

CHAPTER 2

Review of Related Literature and Studies

This chapter supports the conceptual viability of the project by reviewing relevant literature. It aims to identify the key components of the study and provide a synthesis of how the developed project aligns with current literature.

Customer Support Systems and Hybrid Communication

Customer support is a critical factor in determining service quality and customer satisfaction. Ticket-based customer support systems enable organizations to manage inquiries efficiently by organizing requests into trackable tickets (Zendesk, 2023). This structured approach ensures that customer concerns are documented, monitored, and resolved in a systematic manner. In addition, hybrid customer support, which combines digital and in-person communication, has gained importance in modern service environments. Customers prefer multiple communication channels, including online platforms and face-to-face interactions (Salesforce, 2023). Providing both options allows businesses to respond to customer needs more effectively and maintain consistent communication (Zendesk et al., 2023).

In the context of automotive service businesses, hybrid support allows customers to submit inquiries online while still receiving direct assistance during on-site service visits. This supports the integration of a ticket-based hybrid customer support feature in AutoProject+, which aims to organize communication and maintain service records. This approach ensures a high level of responsiveness and overall customer satisfaction

(Salesforce et al., 2023). Furthermore, the shift toward omnichannel support systems highlights the necessity of maintaining context across different interaction points. When a customer initiates a query online, the service staff must have immediate access to that history during a physical visit to avoid repetitive information gathering.

The implementation of ticket-based systems also facilitates better internal resource management. By categorizing tickets based on urgency or service type, staff can prioritize tasks effectively. According to Help Scout (2024), structured digital communication reduces the "mental load" on employees, allowing them to provide more empathetic and accurate responses. This is particularly vital in specialized customization, where technical details are complex and easily misunderstood through verbal-only communication. Digital ticketing acts as a permanent record of the agreement between the garage and the client.

Moreover, the use of automated "canned responses" for common inquiries can drastically improve the speed of service. While the core support remains hybrid, having an automated acknowledgment system informs the customer that their concern has been logged. Research by Freshworks (2023) indicates that the speed of the initial response is a primary driver of trust in digital service platforms. For AutoProject-D, this means the system can manage expectations even during off-hours, reinforcing the reliability of the business.

Hybrid communication also serves as a bridge for the "digital divide" among customers. While younger, tech-savvy clients may prefer pure digital interactions, older or more traditional clients might value the face-to-face consultation. A hybrid model does not force a single path but instead offers a spectrum of engagement. This inclusivity is essential for a localized business in the Philippines, where personal relationships often drive commercial loyalty. By providing a digital ticket for an in-person conversation, the system formalizes a traditional bond.

Technically, the integration of these systems requires a robust backend architecture that can sync real-time data. The use of relational databases allows for the linking of support tickets directly to specific job orders or vehicle profiles. This interconnectedness ensures that the support staff is not working in a silo but has a full 360-degree view of the customer's journey. According to Intercom (2023), data-driven support is 40% more likely to result in first-contact resolution.

In terms of organizational transparency, ticketing systems provide the administrator with oversight on staff performance. Metrics such as "average response time" and "ticket resolution rate" can be generated to identify bottlenecks in communication. This oversight is a key component of modernizing a manual garage, where communication failures often go unnoticed until a customer is lost. Digital support transforms subjective feedback into objective data for business improvement.

The psychological impact of hybrid support on the consumer cannot be overlooked. Knowing that there is a digital "paper trail" for their expensive customization project provides a sense of security. In a high-value industry like automotive modification, the perceived risk is high. Structured support reduces this perceived risk by providing clarity and constant updates. This aligns with the "Assurance" dimension of the SERVQUAL model discussed later in this chapter.

Furthermore, hybrid systems allow for better documentation of "scope creep" in customization projects. When a client requests changes during a project, these can be logged as follow-up tickets rather than verbal asides that might be forgotten or mispriced.

This technical documentation protects both the garage from unpaid labor and the customer from unexpected charges. It creates a professional environment where changes are handled systematically.

Finally, the long-term data gathered from support tickets can inform the garage's service offerings. If a high volume of tickets asks about a specific paint brand or turbocharger kit, the management can adjust their inventory or marketing accordingly. This makes the customer support system not just a reactive tool but a proactive source of business intelligence. AutoProject+ leverages this by centralizing all inquiries into a readable format for the administrator.

Online Appointment and Booking Systems

Online booking systems have become essential in service-oriented industries. Traditional appointment methods often result in scheduling conflicts, delays, and miscommunication due to manual coordination (Ampuan, 2022). These limitations affect both service providers and customers, leading to inefficiencies in daily operations (Ampuan et al., 2022). Modern customers expect fast and convenient booking processes. Digital booking systems allow users to schedule services, receive confirmations, and access updates without the need for direct interaction (Muhammad, 2024).

For automotive repair shops, implementing an online booking system can reduce overlapping schedules and improve workflow organization (Stukalina, 2024). This supports the inclusion of a booking module in AutoProject+, which is designed to manage service appointments in a structured and organized manner (Muhammad et al., 2024). The transition to digital booking allows for a "buffer-based" scheduling logic, where the system automatically accounts for the duration of specific tasks. This prevents a common issue in manual garages where too many complex jobs are accepted for the same day.

A critical advantage of digital booking is the reduction of "no-shows" through automated reminders. According to research by Acuity Scheduling (2023), automated SMS or email reminders can reduce missed appointments by up to 50%. In the automotive sector, where an empty service bay represents lost revenue, this efficiency is vital. AutoProject+ integrates this feature to ensure that the garage's manpower and workspace are utilized at maximum capacity throughout the business week.

Furthermore, online booking platforms allow for 24/7 accessibility. Traditional garages only accept bookings during business hours, but many customers research services in the evening. By providing an always-on booking portal, AutoProject-D can capture leads that might otherwise go to competitors. This aligns with the "Responsiveness" dimension of service quality, ensuring the garage is "open" for inquiries even when the physical shop is closed.

Technically, the booking system must be synchronized with a centralized database to prevent double-booking. The use of a "status-driven" calendar ensures that once a slot is reserved and a fee is paid, that time is immediately blocked for all other users. According to Google Cloud (2024), real-time synchronization is the backbone of modern appointment software. This reliability is what builds long-term customer confidence in the digital platform.

The system also allows for the pre-collection of vehicle data. By requiring the customer to input their car model, engine type, and specific modification goals during the booking process, the garage staff can prepare the necessary tools and parts before the vehicle even arrives. This "pre-arrival" preparation significantly reduces the total service time. It transforms the garage from a reactive workshop into a proactive service center.

Moreover, online booking facilitates better financial planning through reservation fees. By requiring a down payment via GCash or Maya, the garage secures a commitment from the customer. This revenue can be used to purchase specialized parts or materials required for the customization project. This financial workflow is integrated into AutoProject+ to ensure that customization work—which is often capital-intensive—is properly funded from the start.

In terms of user experience, a clean booking interface acts as a "digital storefront." It allows the garage to showcase its organized approach to business. A study by HubSpot (2023) found that 67% of consumers prefer to book appointments through a website rather than a phone call. This preference is even stronger in niche markets like automotive customization, where the target demographic is often younger and highly engaged with digital technology.

From a management perspective, the booking system provides a historical record of "peak times" and "slow periods." The administrator can use this data to plan promotions during slow weeks or hire extra staff during busy seasons. This data-driven management is a far cry from the "guesswork" involved in manual scheduling. It allows for a more sustainable and predictable business model for small garages like AutoProject-D.

Lastly, the integration of role-based access ensures that only authorized staff can modify the schedule. This prevents unauthorized changes that could lead to internal confusion. In AutoProject+, the Staff can view and encode bookings, but only the

Administrator can approve or finalize the calendar. This hierarchy maintains operational discipline and ensures that the garage owner has the final say in the service workflow.

Digital Transformation in Service Management

Digital transformation has significantly influenced the operations of service-oriented businesses. Digital transformation is the integration of digital technologies into business processes to improve operational performance, customer experience, and overall efficiency. Organizations are increasingly adopting digital platforms to remain competitive and meet evolving customer expectations (Vial, 2021). Similarly, digital transformation is no longer optional but a strategic requirement. Businesses that do not adopt digital solutions often experience inefficiencies and reduced competitiveness.

In service industries, digital tools such as web-based systems support automation, reduce manual errors, and improve coordination. In automotive service environments, digital transformation supports the organization of booking processes, job tracking, and customer communication. This highlights the importance of developing systems like AutoProject+ to modernize traditional garage operations (Agustian, 2023). Transformation is not merely about replacing paper with computers; it is about rethinking the entire value chain to provide better service.

The primary hurdle for many small garages is the "legacy mindset" of manual operations. Digital transformation requires a cultural shift where data is seen as an asset. According to McKinsey (2023), small businesses that embrace digitalization see an average productivity increase of 25%. For AutoProject-D, this means the ability to handle more customization projects simultaneously without a corresponding increase in administrative staff.

Technically, this transformation involves the use of "cloud-native" architectures. By hosting AutoProject+ on a web server, the data is accessible from anywhere. The garage owner can monitor revenue while at home, and the mechanic can update job statuses on a tablet. This mobility is a core feature of modern digital management. It breaks the "tether" to a physical desk and a physical logbook, allowing the business to be managed dynamically.

Furthermore, digitalization enables the integration of disparate business functions. Booking, cost estimation, and payment tracking are no longer separate tasks but a unified data flow. This "integration" reduces the risk of data silos where information is lost between different staff members. Research by Deloitte (2024) suggests that integrated systems are 60% less prone to human error than fragmented manual processes.

Customer expectations are a major driver of this change. In an era of instant gratification, customers want to see their vehicle's progress on their phones. Digital transformation allows AutoProject-D to meet this "expectation of visibility." By providing a digital dashboard for the customer, the garage creates a sense of involvement and transparency that is impossible with a manual system. This transparency is a competitive advantage in the trust-sensitive automotive market.

Digital transformation also facilitates better compliance with data privacy standards. While a physical logbook can be easily stolen or lost, a web-based system like AutoProject+ can implement secure encryption and role-based access. This ensures that customer vehicle data and contact information are protected. As data privacy laws become stricter, having a secure digital platform becomes a legal necessity for small businesses.

From a marketing perspective, a digitalized business is easier to promote. The data gathered by the system can be used to send targeted promotions. For example, the system could notify customers who haven't visited in six months about a new "paint protection" service. This "data-driven marketing" is only possible once the core business operations are digitalized. It turns the management system into a growth engine.

Additionally, transformation allows for the use of "analytical engines" to monitor business health. The administrator can see "customer trust scores" or "service completion rates" at a glance. This high-level overview allows for "management by exception," where the owner only needs to intervene when the data shows a problem. This is far more efficient than the constant manual supervision required in traditional garages.

Finally, digital transformation prepares the garage for future technological shifts. As vehicles become more complex and integrated with AI, the management systems must also be capable of handling more advanced data. AutoProject+ serves as the "foundation" for this future. By modernizing now, AutoProject-D ensures it is not left behind by the rapid pace of automotive and technological innovation.

Automated Cost Estimation Systems

Cost estimation is an important component in service industries, particularly in automotive repair and customization. Automated cost estimation systems improve pricing accuracy by using predefined parameters and structured data inputs. These systems help reduce inconsistencies and provide clear pricing details for customers (Alshamrani et al., 2022). Automated estimation tools allow customers to understand expected costs before service execution. This helps build trust and minimizes misunderstandings between customers and service providers.

In the automotive context, cost estimation systems can generate preliminary quotations based on vehicle type, selected services, and labor requirements. This supports the implementation of an automated cost estimation feature in AutoProject+, which aims to provide consistent and reliable pricing information (Choi et al., 2021). The "black box" of manual pricing—where a mechanic quotes a price based on their mood or the customer's appearance—is replaced by a transparent mathematical formula.

The technical logic of an automated estimator relies on a "service-parts-labor" matrix. Each service, such as a "Custom Exhaust Piping," has a base labor rate and a variable material cost. By automating this calculation, the system ensures that the garage maintains its profit margins while the customer receives a fair price. According to AutoData (2024), automated pricing systems can increase shop profitability by 15% simply by ensuring no parts or labor hours go unbilled.

Furthermore, automated estimation reduces the time required to close a sale. In a manual setup, the owner might take hours to calculate a complex quote for a full custom build. With AutoProject+, the client can select their options and receive a preliminary quote in seconds. This speed "captures" the customer's intent while they are still highly motivated, leading to higher conversion rates for the garage.

Accuracy in estimation is particularly critical for high-value customization. A mistake of 10% on a ₱45,000 paint job is a significant loss. By using structured data inputs, the system eliminates "mental math" errors. Research by the Society of Automotive Engineers (SAE, 2023) indicates that digital estimation tools reduce pricing discrepancies by up to 80% compared to manual handwritten quotes.

Automated systems also allow for the generation of "tiered pricing." The garage can offer different levels of service—such as "Standard" vs "Premium" paint—and the system immediately shows the cost difference. This allows the customer to make informed decisions based on their budget. It moves the conversation from "How much will this cost?" to "Which version of this service fits my budget?"—a much more productive sales dialogue.

Technically, the cost estimation module must be "dynamic." As the price of automotive paint or stainless steel rises, the administrator only needs to update the "base rate" in the settings, and all future quotes will be automatically adjusted. This agility is essential in the current economy, where material costs are volatile. It ensures the garage's pricing stays competitive without the need to manually recalculate every service price one by one.

Moreover, the "Preliminary Quote" acts as a legal and operational disclaimer. The system can state that the quote is subject to physical inspection. This protects the garage from unseen damage while still giving the customer a realistic ballpark figure. This balance between "digital speed" and "physical validation" is a key design feature of AutoProject+ to address the complexities of customization.

From a psychological standpoint, a digital quote feels more "official" to the customer. It carries the authority of the system rather than the word of an individual. This reduces the urge for the customer to "haggle" or negotiate, as the price is seen as being generated by a standardized business process. For a small garage like AutoProject-D, this professionalism helps them compete with larger, franchised service centers.

Finally, the data from generated quotes can be analyzed to see which services are most popular but have the lowest "closing rate." If many people request a quote for a turbocharger installation but few book it, the administrator might realize the price is too high or they need a different supplier. This "quote-to-booking" analysis is a powerful tool for optimizing the garage's service menu.

Modernization of Automotive Service Operations

Despite advancements in technology, many automotive repair businesses continue to rely on manual processes. Small and medium-sized garages are under increasing pressure to modernize due to advancements in vehicle technology and customer expectations (Orderry, 2025). Many service shops still depend on traditional booking methods such as phone calls and walk-in transactions (Alev Digital, 2025). While social media platforms provide alternative communication channels, they often lack structured features for managing service records and customer data (Alev Digital et al., 2025).

Scheduling errors, including overlapping bookings and missed appointments, can disrupt operations and result in revenue loss. These challenges highlight the need for a structured digital system that can organize scheduling, communication, and service documentation, which is addressed by the proposed AutoProject+ system (Garage360 et al., 2025). Modernization is about bringing "industrial-grade" management to the local shop level. It is a transition from an "artisan" mindset to a "process-oriented" mindset.

The reliance on social media—while good for marketing—is a major operational risk. When a customer books via Facebook Messenger, that data is trapped in a chat history. It isn't connected to the inventory, the mechanic's schedule, or the financial records. According to research by the Automotive Aftermarket Industry Association (AAIA, 2023), garages using "chat-only" booking lose 20% of their scheduling efficiency due to the time spent scrolling through messages to find details.

Modernization also involves the professionalization of the mechanic's role. Instead of waiting for verbal orders, they log into a digital dashboard. This "Mechanic Module" empowers the staff with information, reducing the constant need to ask the owner for instructions. Research by IMI (Institute of the Motor Industry, 2024) shows that shops with digital task management see a 30% reduction in "idle time" per vehicle.

Paperwork is the enemy of modern efficiency. In a manual garage, a lost job order means a lost car history. By digitalizing these records, AutoProject-D ensures that every car has a "digital twin" in the database. When the car returns a year later, the mechanic knows exactly what paint was used or what pipes were installed. This memory is what allows for high-quality, specialized customization work.

Furthermore, modernization improves the "customer journey." From the first click on the booking site to the final payment confirmation, every step is choreographed. This "experience design" is what differentiates a premium customization shop from a standard repair garage. Modernization allows AutoProject-D to command higher prices by offering a higher level of professional service.

Technically, modernization requires a shift toward "data-driven" decision-making. Instead of "feeling" like the shop is busy, the owner can see the "capacity utilization" percentage. This clarity is essential for scaling the business. If the data shows they are consistently at 100% capacity, it is an objective signal to expand the shop or hire more mechanics.

In terms of local impact, modernization contributes to the professionalization of the Filipino automotive service industry. By adopting global standards in software and management, small shops in Quezon Province can offer world-class service quality.

AutoProject+ is a localized solution that proves digital tools are accessible to even small enterprises with limited budgets.

Moreover, a modernized shop is more resilient to external disruptions. During a health crisis or weather event, a digital business can continue to communicate with its customers and manage its backlog remotely. The "agility" provided by AutoProject+ ensures that AutoProject-D is not a fragile business dependent on a single physical logbook.

Lastly, modernization acts as a legacy-building tool. A business with structured data, a solid customer base, and digital processes is more valuable than one that exists only in the owner's head. It allows the business to be more easily passed on to the next generation or a new manager. AutoProject+ provides the "infrastructure" for the long-term survival of AutoProject-D Custom Garage.

Web-Based Service Management Systems

Web-based systems play a significant role in improving service management and operational efficiency. Web-based applications allow centralized data management, enabling organizations to store, process, and retrieve information efficiently (Peras et al., 2024). These systems also provide accessibility, allowing users to interact with services at any time and from any location (Peras et al., 2024). The web-based systems reduce reliance on manual processes and improve data accuracy.

By automating administrative tasks such as scheduling and record-keeping, organizations can minimize errors and improve productivity (Aziz, 2024). For automotive service providers, web-based systems serve as centralized platforms for managing bookings, tracking services, and maintaining customer records. This aligns with the objectives of AutoProject+ in developing a structured and efficient service management solution (Aziz et al., 2024). The "web-native" nature of the system ensures it is platform-independent, working on phones, tablets, and PCs.

A primary technical advantage of a web-based system is "Instant Deployment." Unlike traditional desktop software that needs to be installed on every computer, a web app is updated on the server. When the researchers add a new feature to AutoProject+, it is immediately available to everyone. This ensures the garage is always using the latest, most secure version of the software. According to Gartner (2024), SaaS (Software as a Service) is the preferred model for 90% of new business applications due to this ease of maintenance.

Furthermore, web-based systems allow for "Seamless Integration" with third-party services. AutoProject+ can easily connect with e-wallets like GCash and Maya because they all speak the same "web language." This connectivity is what enables the automated payment confirmation feature. In a desktop-only world, this would require complex and expensive hardware integrations.

Accessibility is the "superpower" of web systems. The garage owner can check the "Daily Revenue Report" while at a car show, and the staff can encode a walk-in booking while standing next to the customer's car with a tablet. This mobility ensures that the system "follows the workflow" rather than the workflow being forced to a specific computer desk.

Data security in web-based systems has advanced significantly. By using SSL (Secure Sockets Layer) encryption and cloud-based backups, a web-based AutoProject+ is actually safer than a local PC. If the shop's physical computer breaks or is stolen, the data remains safe in the cloud. Research by AWS (2024) indicates that cloud-stored data is 3 times more likely to be recovered after a hardware failure than local storage.

User experience (UX) design is also more flexible in web environments. The interface can be optimized for "touch" on mobile devices and "keyboard" on desktops. This is vital for mechanics who might have greasy hands and need large, clear "Update Status" buttons on a ruggedized tablet. AutoProject+ is designed with these "on-the-floor" realities in mind.

Moreover, web systems facilitate better "Customer Self-Service." Instead of calling to ask "Is my car done?", the customer can just log in. This reduces the administrative load on the staff, freeing them up to focus on technical work. According to a study by Forrester (2023), self-service portals reduce customer inquiry volume by up to 40%.

Centralized data management also enables "Cross-Device Continuity." A customer can start their booking on a desktop at work and finish it on their phone on the way home. The "state" of the interaction is saved on the server. This frictionless experience is what modern consumers expect and what AutoProject+ delivers.

Finally, the web-based model is more cost-effective for small garages. There is no need for expensive server hardware or an on-site IT team. The system runs on the same internet connection the shop already uses. This "low barrier to entry" is what makes sophisticated management software like AutoProject+ feasible for a specialized local shop like AutoProject-D.

Service Quality and SERVQUAL Model in Digital Systems

Service quality is a key factor in customer satisfaction and business success. The SERVQUAL model developed by Parasuraman et al. (1988) identifies five dimensions of service quality: reliability, responsiveness, assurance, empathy, and tangibles (Sumi, 2021). The SERVQUAL model in digital environments has found that service quality has a significant influence on user satisfaction in online systems (Kabir, 2021). Systems that provide reliable and responsive services are more likely to meet user expectations (Sumi et al., 2021).

Furthermore, integrating SERVQUAL dimensions into system design can improve service delivery and customer experience (Ali, 2021). Features such as real-time updates, clear pricing, and organized communication contribute to better service quality. The proposed AutoProject+ system aligns with the SERVQUAL model by supporting reliability through accurate booking records, responsiveness through online support, assurance through cost estimation, empathy through customer data management, and tangibles through a structured digital interface (Ali et al., 2021).

Reliability is arguably the most critical dimension for AutoProject-D. In the customization world, a "reliable" shop is one that finishes the job on the promised date. By using the Agile Scrum Methodology in development, AutoProject+ ensures that the scheduling engine is robust. The system becomes the "source of truth," eliminating the manual errors that lead to broken promises. Reliability in the software builds reliability in the business.

Responsiveness is addressed through the "Hybrid Customer Support" and automated notifications. When a customer receives an instant SMS that their booking is approved, the garage’s responsiveness score increases. The digital system eliminates the "waiting period" that plagues manual operations. In the customer's mind, a fast system is a professional system.

Assurance is built through the "Automated Cost Estimation." When the system provides a detailed quote, it assures the customer that the pricing is not arbitrary. It builds trust by showing the "math" behind the service. This dimension is vital in the Philippines, where "pricing transparency" is a major concern for car owners visiting independent shops.

Empathy in a digital system might seem contradictory, but it is achieved through "Personalization." By storing vehicle history and customer preferences, AutoProject+ allows the staff to say, "Welcome back, we have your previous paint code on file." This "digital memory" shows the customer that the garage cares about their specific car journey. It turns a transaction into a relationship.

Tangibles in the digital realm are the "UI/UX" and the "Digital Documents." A professional, clean booking site and a well-formatted digital invoice are the modern "tangibles" of the business. They represent the "quality" of the intangible service. Research by Nielsen Norman Group (2024) suggests that users judge the quality of a service provider based on the professionalism of their website.

Technically, these dimensions act as "KPIs" (Key Performance Indicators) for the system's evaluation. The researchers can measure the "gap" between what customers expected from the digital platform and what they perceived. This scientific approach ensures that the development of AutoProject+ is not just about "features" but about "outcomes."

Moreover, the SERVQUAL model helps identify which modules of the system are most critical. For example, if users rank "Responsiveness" as most important, the researchers might prioritize the SMS notification engine. This "theory-to-practice" bridge is what gives the research its academic and practical depth.

Additionally, SERVQUAL helps the garage owner understand the "psychology of the sale." It moves the focus away from just "fixing cars" to "delivering a high-quality service experience." This shift is what allows AutoProject-D to position itself as a premium customization garage. The software is the tool that enables this positioning.

Finally, using a globally recognized model like SERVQUAL ensures that the research is grounded in established academic tradition. It provides a common language for the panelists to evaluate the success of the system. It proves that the BSIT students are not just coding a website but are solving a complex service management problem using scientific principles.

ISO/IEC 25010 Software Quality Standards in System Evaluation

Commented [1]: ISO 25010

The ISO/IEC 25010 Software Quality Model provides a scientific framework for evaluating the performance and effectiveness of software systems through specific quality characteristics. According to the international standard, software quality is categorized into eight dimensions: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability (ISO/IEC, 2011). In the development of service management platforms, these standards serve as a benchmark for ensuring that the system is technically sound and capable of supporting business operations (Ouhbi et al., 2022).

Related literature emphasizes Reliability (RL) as a core pillar of software success, defined as the degree to which a system performs specified functions under stated conditions for a specific period. For a web-based system like AutoProject+, reliability is critical to ensure that the digital booking calendar and cost estimation formulas remain accurate and accessible without data loss or downtime (Al-Qutaish, 2023). Studies on automotive service platforms indicate that high system reliability directly correlates with increased user trust, as both garage personnel and customers depend on the integrity of the digital service records.

Furthermore, Performance Efficiency / Responsiveness (RS), particularly time behavior, is identified as a significant factor in user adoption of digital platforms. A study by Valacich et al. (2022) highlights that the responsiveness of a system—measured by how quickly it processes transactions and generates outputs—is essential for maintaining operational flow. In the context of AutoProject+, responsiveness is demonstrated through the automated generation of cost quotations and the immediate delivery of SMS notifications. Related research suggests that when a system exhibits high responsiveness, it reduces the administrative burden on staff and provides a seamless experience for customers tracking their service status (Ouhbi et al., 2022).

Functional Suitability ensures that the system actually does what the garage needs. It evaluates if the "Mechanic Module" actually helps a mechanic or if it’s just extra work. According to research by IEEE (2023), functional suitability is the "make or break" characteristic for small business software. If the system is too complex for the actual workflow, it will be abandoned. AutoProject+ is built to match the actual "Project-D" workflow exactly.

Usability focuses on the learning curve. For a garage where staff might not be technical experts, the system must be intuitive. ISO 25010 usability testing measures "learnability" and "operability." A good system like AutoProject+ should require minimal training to use. This lowers the "friction" of modernization for the business owner.

Security is a non-negotiable standard in the digital era. With integrated GCash and Maya payments, the system must protect financial data. ISO 25010 security standards evaluate "confidentiality" and "integrity." The researchers must ensure that one customer cannot see another’s private vehicle or payment details. This technical "wall" is a key requirement of the project.

Compatibility ensures the system works across different browsers and devices. Since AutoProject-D staff might use an old PC and a new Android tablet, the system must be "responsive" in its design. According to W3C (2024), compatibility issues are the leading cause of "user drop-off" in web applications. AutoProject+ is tested for cross-browser stability.

Maintainability evaluates how easy it is to update the system. If the garage adds a new service, like "Ceramic Coating," the system should be easy to modify. A "maintainable" system uses clean, documented code. By using the Laravel framework, the researchers ensure that AutoProject+ follows professional coding standards that allow for future expansion.

Finally, the integration of ISO/IEC 25010 standards provides a scientific foundation for evaluating the technical success of AutoProject+. By aligning the system’s development with recognized benchmarks, the researchers ensure that the platform is not only functional but also meets international standards for software quality. This technical evaluation complements the SERVQUAL model by focusing on the system's internal robustness while SERVQUAL focuses on the customer's external perception of service quality.

Cost-Benefit Analysis in Digital Systems

The implementation of web-based management systems in service-oriented businesses provides a strategic framework for improving operational performance and long-term financial sustainability. In the context of automotive service management, digital transformation allows small-scale enterprises to transition from manual record-keeping to automated processes, which significantly reduces administrative overhead and minimizes costly human errors (Keach L, 2025). Integrating features such as automated cost estimation and digital booking, businesses can optimize internal resource utilization and ensure that labor and materials are allocated more efficiently.

This level of automation decreases the dependency on manual labor for routine tasks such as scheduling or manual billing, allowing garage personnel to focus on higher-value activities such as specialized customization and technical quality assessment (Sharma, S., et al., 2021). From an economic perspective, these operational efficiencies translate into tangible cost savings and improved revenue performance. A formal cost-benefit analysis of such systems demonstrates that while there is an initial investment in development and cloud infrastructure, the long-term benefits result in a high return on investment.

One primary "benefit" is the reduction of "Opportunity Cost." When the owner is stuck doing manual paperwork for two hours a day, that is time they are not spending on complex, high-profit customization builds. If their labor is worth ₱500/hour, that is ₱1,000/day in "lost time." AutoProject+ recaptures this time, allowing the business to earn more without working longer hours. According to Harvard Business Review (2023), automation in small businesses pays for itself within the first 12 months.

Furthermore, the "Benefit of Data" is immense. Without a digital system, the garage might not know that they are actually losing money on "Paint Touch-ups" because they underestimated the labor time. A cost-benefit analysis of the system includes the "benefit of accurate pricing." By ensuring every job is profitable, the system prevents the "hidden losses" that sink manual businesses.

Material waste reduction is another tangible benefit. By using the system to track the exact materials used in previous projects, the garage can buy more accurately. In customization, where paint and piping are expensive, reducing waste by 10% through better data can save thousands of pesos per year. This "operational efficiency" is a direct financial outcome of the software.

The "Cost" side of the analysis includes the development time and hosting fees. For a BSIT capstone, the development "cost" is theoretically free for the business, making the ROI (Return on Investment) extremely high. The only ongoing cost is the domain and hosting fee (approximately ₱3,000/year). This minimal investment compared to the potential for increased revenue makes the system economically unassailable.

Customer retention is a "long-term benefit." A customer who has a smooth, professional experience at AutoProject-D is more likely to return and recommend the shop to others. If the system brings in just two extra "Full Custom Build" clients per year (at ₱45,000 each), the system has generated an extra ₱90,000 in revenue. This "growth benefit" far outweighs the initial development efforts.

Moreover, the system reduces the "cost of errors." A double-booking that results in a disgruntled customer who leaves a 1-star review on social media has a real financial cost. It can drive away dozens of potential clients. AutoProject+ acts as a "risk mitigation" tool, protecting the garage’s most valuable asset: its reputation.

The "Intangible Benefits" should also be quantified. The reduced stress on the owner and the improved morale of the mechanics who have clear, digital instructions are vital. While harder to put a peso sign on, these factors contribute to the "longevity" of the business. A business that is easier to run is a business that stays open longer.

Finally, the adoption of a web-based service booking and cost estimation system not only supports improved garage performance but also strengthens the economic sustainability of automotive customization businesses, demonstrating clear advantages in cost-benefit terms (Pronamics et al., 2025). When compared to traditional manual garage operations, a centralized digital platform accelerates the service journey and optimizes resource utilization, which enhances overall business profitability (Lumenalta et al., 2025).

Synthesis

Small automotive businesses face significant challenges because they often rely on manual operations that cannot meet the specific demands of modern consumers. Three key operational factors including scheduling, cost estimation, and communication have a major impact on a garage's efficiency, revenue, and customer trust. As customers increasingly expect real-time updates and data-driven responses, businesses using traditional methods may suffer from overlapping appointments, lost revenue, and customer dissatisfaction.

Manual cost estimation, often performed verbally or through handwritten notes, is especially problematic because it leads to pricing inconsistencies and misunderstandings about the scope of work. Furthermore, when records are not stored digitally, it becomes difficult to review past services or monitor overall business performance. Because of this limitation, a centralized digital framework becomes essential for sustainable growth.

In addition to operational hurdles, traditional communication practices create further difficulties. Most garages still rely heavily on phone calls, walk-ins, and social media, which requires significant manual effort and is prone to errors such as forgotten booking details. Manual scheduling frequently results in idle service slots or unexpected gaps caused by no-shows. These situations disrupt the planned workflow and prevent customers from having a consistent way to track their vehicle's progress. This lack of transparency ultimately reduces accountability among personnel and negatively affects the customer experience.

Because of these challenges, there is a clear need for smarter web-based systems in the automotive service sector. Research indicates that digitalization through platforms like AutoProject+ can significantly improve service management by monitoring booking requests, service status, and revenue in real time. The integration of the ISO/IEC 25010 Software Quality Model provides a scientific foundation for this digitalization by ensuring that the system meets international standards for Reliability (RL) and Performance Efficiency (Responsiveness). Literature suggests that a system’s technical reliability—its ability to maintain data integrity and avoid downtime—is the primary driver of organizational trust. Similarly, the responsiveness of the system in generating quotations and processing SMS notifications is crucial for maintaining a professional service pace.

Integrating the SERVQUAL model further strengthens the system because it allows service quality dimensions to be addressed directly through these technical features. For example, an automated cost estimation system provides the Assurance and Reliability required to minimize human error, while digital job orders allow mechanics to track their progress so that work remains organized and customers stay informed. By combining the technical rigor of ISO/IEC 25010 with the customer-centric focus of SERVQUAL, AutoProject+ creates a comprehensive digital ecosystem that ensures both high-quality software performance and superior automotive service delivery.

CHAPTER 3

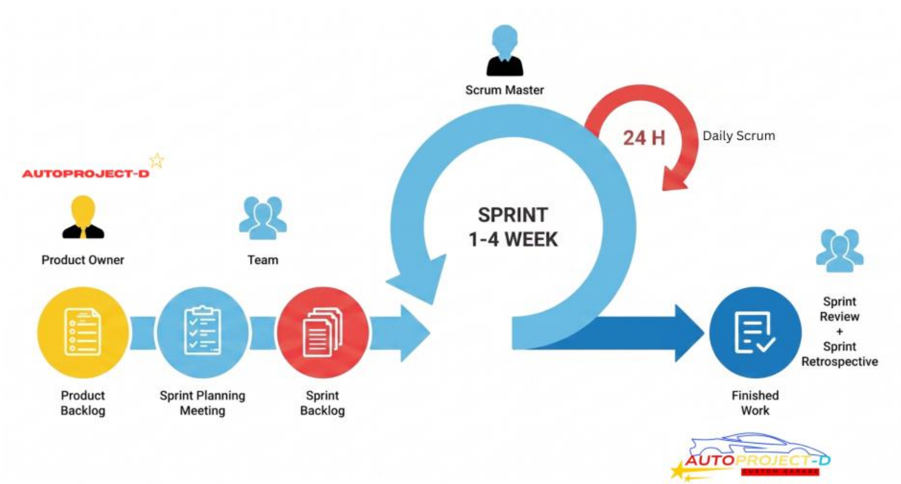
DESIGN AND DEVELOPMENT METHODOLOGY

This chapter presents the research design and methodology used in the development of the system. It details the procedures for gathering data, defining requirements, analyzing user needs, and designing the system architecture. The research instruments used to collect information from stakeholders are also discussed. Furthermore, this chapter outlines the chosen Software Development Life Cycle (SDLC) model, its phases, and how it was applied throughout the system development process.

Software Methodology

Figure 3

Agile Scrum Methodology



This figure illustrates that the researchers adopted Agile Scrum as the primary software development framework to guide the design and implementation of the AutoProject+ system. This methodology is characterized by its iterative approach, focusing on delivering high-quality software through continuous feedback and small, manageable work cycles called Sprints. Adopting this framework is highly advantageous for AutoProject-D Custom Garage because it allows the development team to remain flexible and responsive to the evolving operational needs and complex technical requirements of automotive customization. By breaking the project into iterative loops, the researchers ensure that each module, from cost estimation to payment processing, is refined based on stakeholder input and technical validation.

The process begins with the creation of a Product Backlog, where the researchers define the high-level requirements and domain scope of the garage management system. This is followed by Sprint Planning, a phase where the team identifies core objectives and sequences tasks—such as user authentication and role-based access—to align with the business goals of Mr. Makol Pasumbal. During each Sprint, the researchers focus on the Engineering and Development quadrant, constructing specific modules like the digital booking flow and mechanic interface using PHP and the Laravel framework.

Throughout the development cycle, the team conducts Daily Scrums to monitor progress and address potential technical hurdles, such as the logic required for the automated cost estimation engine. At the conclusion of each Sprint, the system moves into the Evaluation and Review phase, where functional versions of the modules are rigorously tested against ISO 25010 quality standards to ensure high reliability and usability for staff and customers. This iterative cycle ensures that the AutoProject+ platform evolves from a foundational model into a complete, synchronized web-based ecosystem.

The development is organized into several key Sprints that represent the technical evolution of the system:

Sprint 1. System Foundation and Security This initial phase focuses on building the backbone of the platform. The researchers establish secure User Authentication and Role-Based Access Control (RBAC) to distinguish between Customers, Administrators, Staff, and Mechanics. During this time, the User Database (D1) is structured using MySQL to safely store hashed credentials, contact details, and electronic mail for notifications. This foundation ensures that every user has a personalized dashboard and that sensitive data is protected according to ISO 25010 security standards.

Sprint 2. Core Logic and Reservation Flow This sprint centers on the Automated Cost Estimation Engine, which is a critical innovation of the system. The team develops the mathematical logic to process vehicle details and service rates—such as intake piping

(₱6,000–₱8,000) or custom paint jobs (₱15,000–₱20,000)—to generate preliminary digital quotes. Simultaneously, a synchronized digital calendar is implemented to handle booking requests. This sub-process queries the Booking Database (D2) for scheduling conflicts, ensuring that garage capacity and mechanic availability are cross-referenced before a slot is reserved.

Sprint 3. Technical Execution and Payment Integration During this phase, the focus shifts to the garage floor and financial accountability. The researchers build the Mechanic Module, which features a status-driven queue where jobs are tracked from "Queued" to "Completed". This allows mechanics to provide real-time updates and add technical remarks directly to the Service Database (D3). To finalize bookings and reduce no-shows, the system is integrated with GCash and Maya payment gateways. This allows for the secure processing of reservation fees, which are manually validated by the Administrator to maintain strict financial control.

Sprint 4. Hybrid Support and Business Intelligence The final sprint implements tools for long-term engagement and management oversight. The Hybrid Customer Support system is developed, allowing customers to submit inquiries via Support Tickets (D5) that are monitored and resolved by the staff. Additionally, the team integrates a Business Analytics module that draws data from the Payment Records (D4) and job history to generate automated performance reports. These reports, which can be exported as PDFs, provide the Administrator with insights into revenue trends, mechanic productivity, and the most requested services, supporting data-driven management decisions.

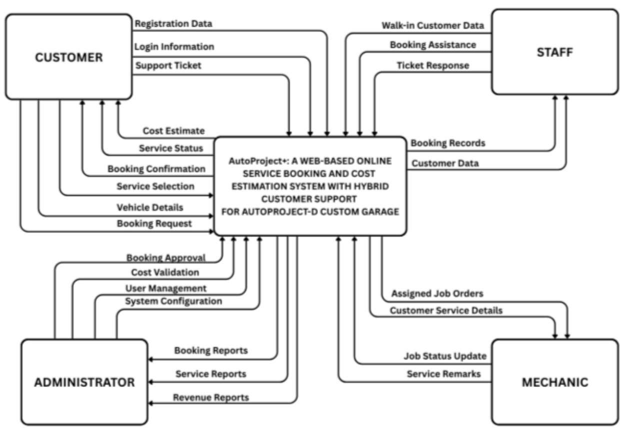
System and Data Modeling

This section provides an in-depth analysis of the data and processing elements crucial to flow and handling of data throughout the system. Each diagram has a distinct purpose, helping to illustrate how the system functions within its environment clearly.

Context Diagram

Figure 4

AutoProject+ Context Diagram



The context diagram illustrates the functional boundaries of the AutoProject+ system and its interaction with four primary external entities: the Registered Client, the Administrator, the Staff, and the Mechanic. The system serves as a centralized repository that facilitates the transition from manual garage operations to a structured digital workflow. Unlike automated payment systems, the prototype utilizes a manual verification process for financial transactions to ensure accuracy and administrative control over the garage's e-wallet accounts.

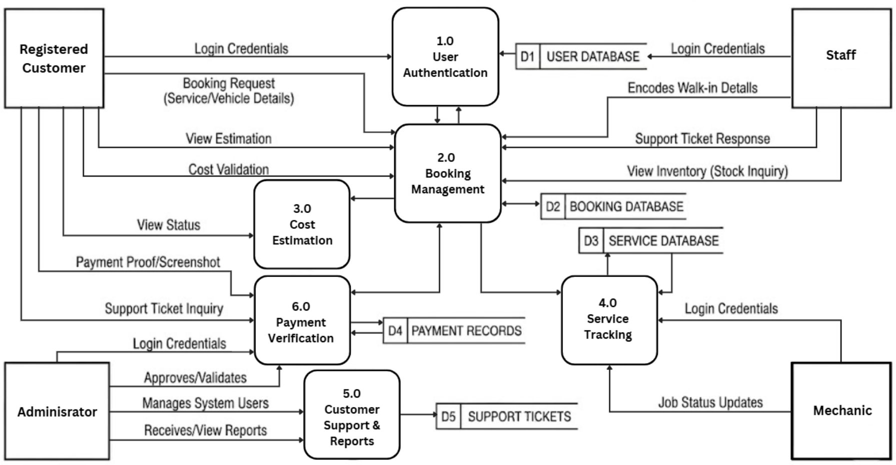
The Registered Client acts as a primary external entity that provides registration data, vehicle specifications, and service requests to the platform. To formalize a booking, the client submits a Payment Proof or Reference Number directly through the system interface following an e-wallet transaction. The Administrator, identified as the garage owner, receives notification of these pending transactions and validates them against external records. Once verified, the Administrator inputs a Cost Validation into the system, which then triggers a Payment Acknowledgment and Booking Confirmation back to the client.

Internal coordination is managed through the Staff and Mechanic entities. The Staff entity provides walk-in customer data and handles support ticket responses to maintain high levels of responsiveness. The Mechanic entity receives Assigned Job Orders and specific technical details through the platform, providing manual Job Status Updates as the work progresses from the queued stage to completion. This flow of information ensures that all data related to payments, scheduling, and service execution is centralized, reducing the reliance on fragmented paperwork and verbal instructions while maintaining a clear audit trail for the Administrator.

Data Flow Diagram

Figure 5

AutoProject+ Data Flow Diagram (Level 0)



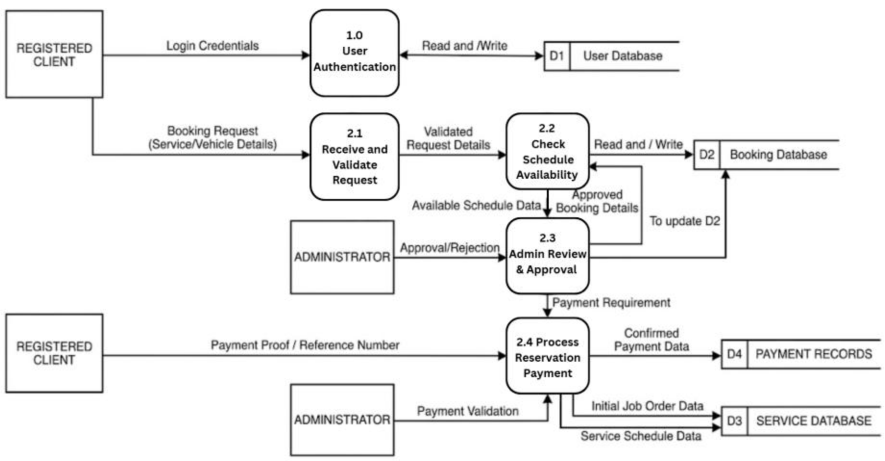
This figure illustrates the Level 0 Data Flow Diagram (DFD) of the AutoProject+ System, providing a high-level overview of the major functional modules and the structured flow of information between external entities and the centralized database. The system's workflow is grounded in a synchronized architecture that ensures all data transitions through dedicated processes before being stored or retrieved.

The cycle begins with User Authentication (1.0), where login credentials from registered clients, staff, mechanics, and administrators are validated against the User Database (D1) to establish role-based access. Once authenticated, the Booking Management (2.0) process centralizes service requests from online reservations and walk-in data encoded by the staff, writing these records into the Booking Database (D2). This process is directly linked to Cost Estimation (3.0), which retrieves service parameters and vehicle details to generate the preliminary quotations necessary for customer assurance.

Operational transparency is maintained through the Service Tracking (4.0) module, where mechanics input manual job status updates and service remarks. These updates are recorded in the Service Database (D3), allowing the system to provide real-time visibility to both customers and administrators. Financial accountability is handled through the Payment Verification (6.0) process, where clients submit payment proof or reference numbers. The administrator validates these entries, and upon approval, the system updates the Payment Records (D4). Finally, the Customer Support and Reports (5.0) process manages the exchange of Support Tickets (D5) and allows the administrator to generate operational summaries for revenue and service performance. This DFD architecture ensures that every data flow follows recognized standards, focusing on technical reliability, professional documentation, and administrative oversight.

Figure 6

AutoProject+ Data Flow Diagram (Level 1)



This figure illustrates the Level 1 Data Flow Diagram (DFD), which provides a detailed decomposition of the Booking Management (2.0) process. By breaking down the main process into specific sub-processes, the diagram clarifies how the system handles the transition from an initial customer inquiry to a finalized, paid reservation.

The workflow begins with Receive and Validate Request (2.1), where the system captures the Registered Client's vehicle details and customization selections. Once the request is structured, it moves to Check Schedule Availability (2.2). This process performs a "Read and Write" operation with the Booking Database (D2) to ensure there are no overlapping appointments or capacity issues at the garage.

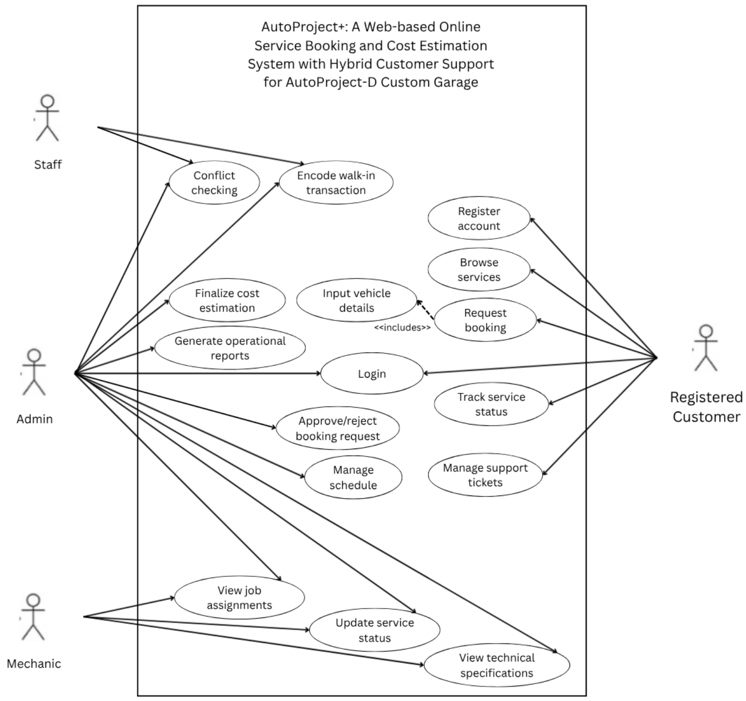
Following availability confirmation, the data flows to Admin Review and Approval (2.3). Here, the Administrator acts as a critical external entity, providing either an "Approval" or "Rejection" based on the feasibility of the customization. Upon approval, the system triggers a Payment Requirement, which leads to the Process Reservation Payment (2.4) module.

In this final sub-process, the Registered Client submits a Payment Proof or Reference Number following an e-wallet transaction. The Administrator performs a manual Payment Validation within the system to ensure the funds are received. Once verified, the process simultaneously updates the Payment Records (D4) and the Service Database (D3), which generates the Initial Job Order Data required for the mechanics to begin work. This granular decomposition demonstrates how the system architecture eliminates manual scheduling errors and ensures financial accountability before any physical resources are allocated to a project.

Use Case Diagram

Figure 7

AutoProject+ Use Case Diagram



The use case diagram illustrates the functional requirements of the AutoProject+ system by defining the interactions between four primary actors and the system boundaries. The diagram is structured in a portrait format to clearly delineate the roles of the Registered Client and Staff on the left, and the Administrator and Mechanic on the right. This architectural layout ensures that every user goal is mapped to a specific system function, facilitating a role-based access control (RBAC) environment.

The Registered Client initiates the service lifecycle through the Register and Login use cases. Their primary goals include browsing the catalog of customization services and initiating a Request Booking, which technically includes the Input of Vehicle Details. To formalize a reservation, the client interacts with the Submit Reservation Payment use case; this is an extend relationship, meaning it is an optional flow triggered after the booking is requested. This payment process is further extended by specific options like Upload GCash/Maya Screenshot or Request In-Person Payment, reflecting the prototype's manual verification logic. Additionally, the client can utilize the Track Service Status and Manage Support Tickets functions to maintain visibility over their vehicle's progress.

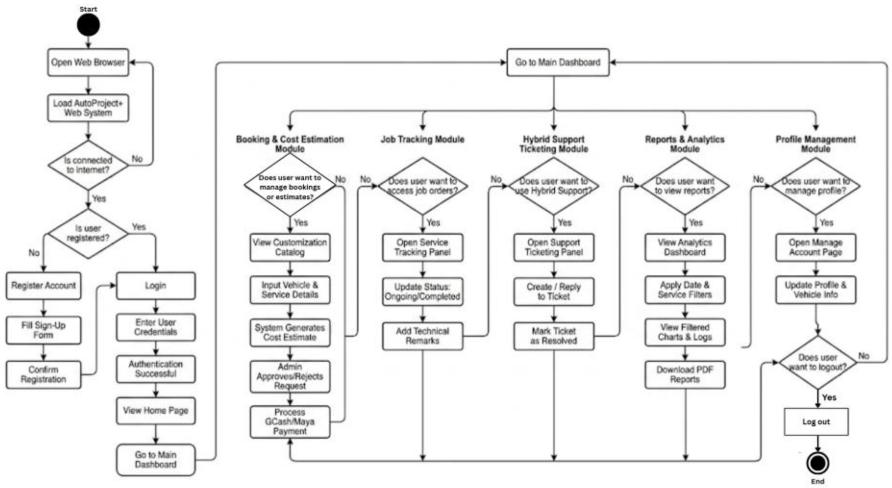
On the operational side, the Staff actor is responsible for Encoding Walk-in Transactions to ensure that customers without internet access are still integrated into the digital workflow. The Staff also interacts with the Manage Schedule use case, which includes a Conflict Checking sub-process to prevent overlapping appointments, a direct solution to the garage's previous scheduling issues. Meanwhile, the Mechanic interacts specifically with the Update Service Status and View Technical Specifications use cases, providing the "on-the-floor" data that feeds back into the client's tracking dashboard.

The Administrator possesses comprehensive oversight, accessing the system via an Admin Login that includes Audit Trail Logging and an Analytics Engine for security and data integrity. The Administrator's use cases focus on high-level management, such as Finalizing Cost Estimations, Approving or Rejecting Booking Requests, and Generating Operational Reports. By structuring the system this way, AutoProject+ ensures that every action is documented and role-specific, transforming the traditional, fragmented operations of AutoProject-D into a modernized, transparent, and technically sound automotive service platform.

Activity Diagram

Figure 8

Activity Diagram of AutoProject+



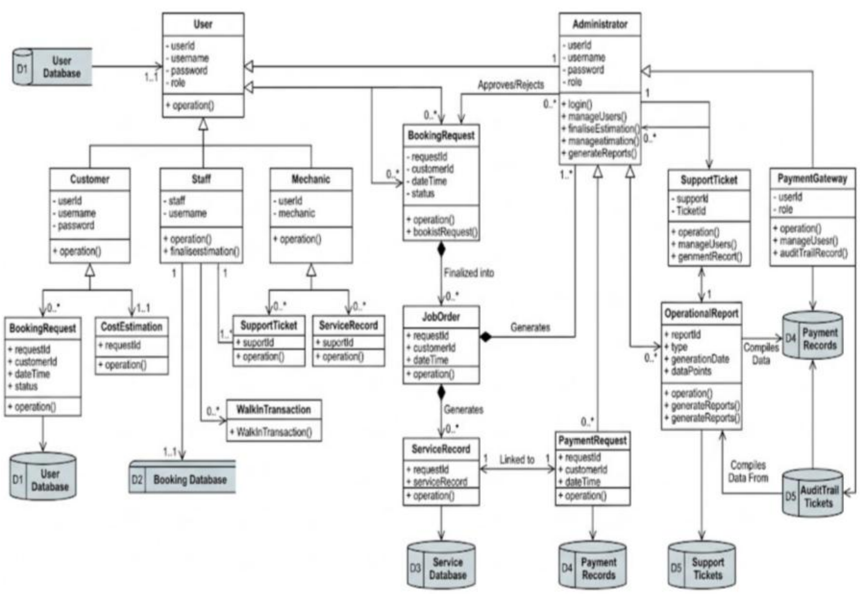
The system flowchart illustrates the step-by-step operational flow of the AutoProject+ Web System from start to finish. It begins when the user opens a web browser and loads the application. After verifying active internet connectivity, the system checks the user's registration status. New users are routed to register an account, fill out a sign-up form, and confirm their registration, while existing users proceed directly to the login phase. Upon successfully entering their credentials and completing authentication, users are directed to the home page and subsequently routed to the main dashboard. From this central hub, users can access a variety of specific functions based on their current needs. The system navigates through distinct decision-driven modules, allowing users to manage bookings and process cost estimates with GCash or Maya payment integration, access job orders to update service tracking statuses and technical remarks, or utilize the hybrid support module to create and resolve support tickets. Users can also access the reports and analytics module to apply data filters and download PDF summaries, or navigate to the profile management module to update their personal and vehicle information. This logical progression ensures that all platform features are easily accessible, providing a streamlined and efficient experience from initial login until the user decides to log out and terminate the session.

As shown in Figure 8, the architectural diagram of the AutoProject+ Web System illustrates the underlying structure of the platform, representing the relationships, database entities, and data flow within the application. The diagram maps the dynamic interactions between the user interface, the administrative backend, and the various integrated modules managing the automotive service workflows.

Class Diagram

Figure 9

AutoProject+ Class Diagram



This figure illustrates the Class Diagram for the AutoProject+ System and provides a detailed view of the system's static structure, including its object classes, attributes, operations, and the logical relationships between them. The diagram shows how the system applies Object-Oriented Programming (OOP) principles such as inheritance and encapsulation to manage complex automotive service data.

The diagram includes a base User class that contains common attributes such as userId, username, and password. This class is extended through inheritance into four subclasses: Customer, Staff, Mechanic, and Administrator. Each subclass represents a specific role in the system and includes its own operations and access privileges according to its responsibilities.

For service and booking management, the BookingRequest class stores the initial information submitted by customers when requesting a service. After the request is reviewed and approved by the administrator, it is converted into a JobOrder. The JobOrder class serves as the central component that manages the service workflow and generates ServiceRecord objects, which document the customization or repair work carried out by mechanics.

The system also includes a CostEstimation class that is associated with the booking process. This class generates preliminary price estimates based on selected services and vehicle information. For financial transactions, the PaymentGateway class manages interactions between PaymentRequest and PaymentRecords classes to support digital payments for reservation fees.

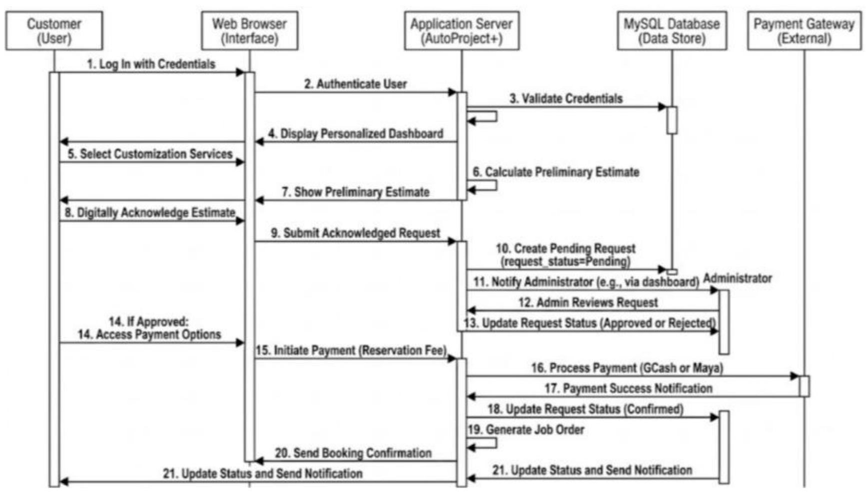
Customer support and administrative reporting are also represented in the diagram. The SupportTicket class enables the hybrid support feature, allowing customers to communicate with staff regarding inquiries or issues. Meanwhile, the OperationalReport class collects and processes data from different system components such as payment records and job orders in order to generate summarized reports for the administrator.

All classes are connected to their respective database tables to ensure proper data storage and retrieval. These include the User Database (D1), Booking Database (D2), Service Database (D3), Payment Records (D4), and Support Tickets (D5). This organized class architecture ensures that the AutoProject-D Custom Garage system operates on a modular and scalable platform while maintaining data integrity through clearly defined relationships and responsibilities.

Sequence Diagram

Figure 10

AutoProject+ Sequence Diagram



This figure illustrates the Sequence Diagram for the AutoProject+ System, detailing the specific, time-ordered interactions between various system objects and external actors during the Booking Request and Confirmation process. This diagram provides a dynamic view of how the system fulfills a user's goal by showing the sequential flow of messages.

The diagram follows the journey of a customer booking from initial login to final confirmation, organized vertically by time and horizontally by object/actor lifelines:

Phase 1: Authentication and Service Selection (Items 1-5)

The process begins with the Customer (User) interacting with the Web Browser (Interface) to log in. The interface sends authentication credentials to the Application Server (AutoProject+), which verifies the user against the MySQL Database (Data Store). Once authenticated, the server returns a personalized dashboard, and the customer selects the desired customization services.

Phase 2: Estimate Generation and Administrative Approval (Items 6-13)

The Application Server performs an internal operation (6. Calculate Preliminary Estimate) and displays it to the customer. The customer must then 8. Digitally Acknowledge Estimate. Upon submission, the server 10. Creates Pending Request in the database and 11. Notifies Administrator. The system then awaits administrative intervention. The Administrator (Actor) reviews the request via the interface, and updates the request status (13. Update Request Status) based on criteria like garage availability.

Phase 3: Payment and Job Order Generation (Items 14-20 - If Approved Branch)

If the status is "Approved," the customer accesses payment options. When the customer initiates payment for the reservation fee, the application server integrates with the external Payment Gateway (e.g., GCash or Maya). The gateway processes the transaction and provides a 17. Payment Success Notification. The system then 18. Updates Request Status (Confirmed) and automatically 19. Generates Job Order. Finally, the customer receives a booking confirmation.

Alternative Path: Rejection (Items 21)

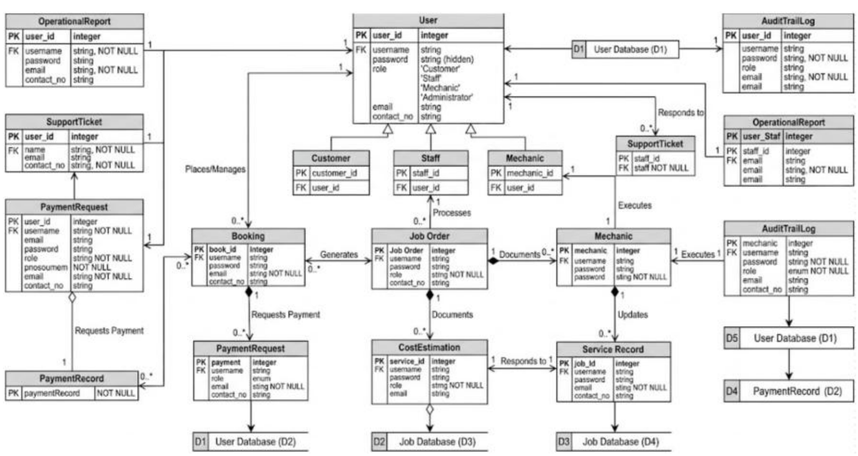
If the Administrator rejects the request in Step 13, the Application Server triggers an automated alternative flow, immediately notifying the database and interface, which in turn informs the customer (21. Update Status and Send Notification), effectively ending the process.

This sequence diagram emphasizes the interconnected nature of the AutoProject+ objects, illustrating exactly which components are responsible for processing data, validating security, managing payments, and maintaining digital traceability throughout the booking lifecycle.

Entity Relationship Diagram

Figure 11

AutoProject+ ERD



This figure illustrates the Entity Relationship Diagram (ERD) for the AutoProject+ System. It presents the database schema and the logical relationships among the different data entities within the system. The ERD serves as the foundation of the system's data architecture, ensuring that information related to users, services, and transactions is organized, consistent, and stored with high integrity. The design uses Primary Keys (PK) and Foreign Keys (FK) to establish clear relationships between entities and to minimize data redundancy.

The database structure begins with the User table, which stores essential account credentials and identification details. This table is connected to several role-specific tables, including Customer, Staff, Mechanic, and Administrator. This design allows the system to maintain a single identity record for each individual while still storing role-specific information. For example, a mechanic may have specialization details recorded, while a customer may have contact information and service history associated with their account.

The booking and transactional process starts when a customer submits a service request, which creates a record in the Booking table. This table is connected to the PaymentRequest table through a one-to-many relationship so that financial transactions can be associated with a particular service request. The CostEstimation table is also connected to the booking records to store pricing estimates that were generated during the early stage of the request.

After a booking is approved, it produces a JobOrder, which becomes the main operational record used inside the garage. The JobOrder table is associated with the Mechanic who performs the work and the ServiceRecord that documents the tasks completed on the vehicle. This structure allows administrators and staff to monitor which mechanic handled a specific job and what services were performed for each vehicle customization project.

The system also supports communication and monitoring features. The SupportTicket table links users with staff members, allowing service concerns or inquiries to be recorded and managed within the system. In addition, an AuditTrailLog records important system events and data modifications to maintain accountability and provide administrative oversight.

The ERD also represents how system data is organized into several logical storage groups. These include the User Database (D1), Booking Database (D2), Service Database (D3), Payment Records (D4), and Support Tickets (D5). Organizing data into these logical clusters improves system performance and allows sensitive information such as financial records and personal data to be protected with appropriate security measures.

This Entity Relationship Diagram highlights the strong relational structure of the AutoProject+ System. From the moment a customer registers and submits a booking request up to the completion of services and the generation of financial reports, all data entities are interconnected. This structure supports efficient data management, accurate reporting, and reliable operational tracking within the AutoProject+ platform.

Data Dictionary

Data dictionary serves as a comprehensive catalog that defines the names, descriptions, types, and attributes of all data elements stored within a system's database. The following tables present the specific data dictionary designed for the AutoProject+

system, detailing the underlying data architecture and structural framework that support the platform's core functionalities.

Table 1

Data Dictionary for Users

Field Name	Data Type	Constraint	Description
user_id	INT	PK, Auto-Inc	Unique identifier for each user account.
username	VARCHAR(50)	Unique, NOT NULL	The name used for system login.
password	VARCHAR(255)	NOT NULL	Hashed security credential.
role	ENUM	NOT NULL	Defines access level: 'Customer', 'Staff', 'Mechanic', 'Admin'.
email	VARCHAR(100)	Unique, NOT NULL	User's electronic mail for notifications.
contact_no	VARCHAR(15)	NOT NULL	Mobile number for SMS alerts.

Table 1 illustrates the data dictionary for the Users table, which contains six (6) field names with one (1) primary key. These fields store essential authentication and profile information for all individuals accessing the AutoProject+ system, including their unique username, hashed password, designated system role, electronic mail, and contact number. The user_id serves as the primary key that uniquely identifies each account. This table is critical for managing secure logins, maintaining communication channels, and enforcing role-based access control for customers, staff, mechanics, and administrators.

Table 2

Data Dictionary for Bookings

Field Name	Data Type	Constraint	Description
booking_id	INT	PK, Auto-Inc	Unique identifier for the booking request.
customer_id	INT	FK (Users)	References the customer making the request.
service_type	VARCHAR(100)	NOT NULL	Category of customization (e.g., Paint, Engine).
vehicle_details	TEXT	NOT NULL	Specifics like Make, Model, and Plate Number.
status	VARCHAR(20)	Default: 'Pending'	Current state: 'Pending', 'Approved', 'Rejected'.
total_estimate	DECIMAL(10,2)	NOT NULL	The preliminary cost calculated by the system.

Table 2 illustrates the data dictionary for the Bookings table, which contains six (6) field names including one (1) primary key and one (1) foreign key. These fields capture the details of a customer's service request, such as the specific category of customization, detailed vehicle information, current approval status, and the system-generated cost estimate. The booking_id acts as the primary key, while customer_id serves as a foreign key linking the request to a specific user. This table is used to track the lifecycle of an appointment from the initial online inquiry to administrative validation.

Table 3

Data Dictionary for Job Orders

Field Name	Data Type	Constraint	Description
job_order_id	INT	PK, Auto-Inc	Unique identifier for the active work order.
booking_id	INT	FK (Bookings)	Links the work to the original customer request.
mechanic_id	INT	FK (Users)	References the mechanic assigned to the task.
current_status	VARCHAR(20)	NOT NULL	Stages: 'Queued', 'Ongoing', 'Paused', 'Completed'.
start_date	DATETIME	NOT NULL	Timestamp when the actual work began.
remarks	TEXT	Optional	Technical notes added by the mechanic.

Table 3 illustrates the data dictionary for the Job Orders table, which contains six (6) field names with one (1) primary key and two (2) foreign keys. These fields manage the technical execution of a project, recording the assigned mechanic, the real-time workflow stage (such as 'Ongoing' or 'Paused'), the official start date, and specific technical remarks. The job_order_id is the primary key, with booking_id and mechanic_id acting as foreign keys to maintain traceability. This table is essential for monitoring garage productivity and ensuring mechanics have clear digital instructions for every assigned vehicle.

Table 4

Data Dictionary for Payments

Field Name	Data Type	Constraint	Description
payment_id	INT	PK, Auto-Inc	Unique transaction identifier.
booking_id	INT	FK (Bookings)	Links the payment to a specific service.
amount	DECIMAL(10,2)	NOT NULL	Total amount paid (Reservation Fee).
method	VARCHAR(20)	NOT NULL	Payment channel used: 'GCash' or 'Maya'.
reference_no	VARCHAR(50)	Unique	Tracking number provided by the payment gateway.
payment_date	TIMESTAMP	Default: CURRENT	Date and time the transaction was confirmed.

Table 4 illustrates the data dictionary for the Payments table, which contains six (6) field names including one (1) primary key and one (1) foreign key. These fields document the financial transactions of the system, specifically recording the reservation fee amount, the chosen digital payment method (GCash or Maya), the gateway-provided reference number, and the precise timestamp of the transaction. The payment_id serves as the primary key, and booking_id links the payment to the corresponding service request. This table ensures financial accountability and automates the verification of booking deposits.

Table 5

Data Dictionary for Support Tickets

Field Name	Data Type	Constraint	Description
ticket_id	INT	PK, Auto-Inc	Unique identifier for the support inquiry.
customer_id	INT	FK (Users)	References the customer seeking help.
subject	VARCHAR(255)	NOT NULL	Short description of the issue or inquiry.
message	TEXT	NOT NULL	Detailed description provided by the user.
staff_id	INT	FK (Users)	References the staff member handling the ticket.
is_resolved	BOOLEAN	Default: FALSE	Indicates if the issue has been closed.

Table 5 illustrates the data dictionary for the Support Tickets table, which contains six (6) field names with one (1) primary key and two (2) foreign keys. These fields facilitate the hybrid support system by storing the subject of an inquiry, the detailed message from the customer, the assigned staff member, and a boolean status indicating if the issue is resolved. The ticket_id is the primary key, while customer_id and staff_id serve as foreign keys to connect the two parties. This table is used to maintain high responsiveness and provide a documented history of all customer-service interactions.

Methods of Data Gathering

To ensure the development of **AutoProject+** is grounded in actual business needs and industry standards, the following data gathering methods were employed by the researchers:

Qualitative Method. This research method is an approach used to study and interpret complex phenomena based on non-numerical data. The researchers employed a qualitative method to gather in-depth information about the requirements of the developed system. A total of ten (10) respondents, including the owners, staff, and regular clients of AutoProject-D Custom Garage, participated in this method to provide relevant insights, preferences, and feedback regarding the customization process.

Interview. It refers to the collection methods of data that involve direct conversations with individuals. The proponents conducted semi-structured interviews with the owner and the lead mechanic of AutoProject-D Custom Garage to identify the specific hindrances they encounter during the manual booking and project tracking process. These conversations were vital in defining the necessary status updates and the logic for the automated cost estimation engine.

Web Search. It refers to gathering relevant information about the topic of the developed system from the internet. The proponents conducted extensive research across various academic databases and technology blogs to gather reliable articles concerning automotive management systems and digital payment integration. This method helped the researchers understand current market trends in the car customization industry.

Document Review. The researchers gathered information and evidence from existing records to support the study. This included reviewing the garage's manual logbooks, previous service invoices, and existing job order forms. The goal was to gain insights into how data was historically recorded and to ensure that the new digital system maintains the same level of technical detail while improving accessibility and organization.

System Requirements

Functional Requirements

A functional requirement describes the specific operations and functions that the proposed system must perform to meet the needs of its users. It defines how the system should behave, including the inputs it receives, the processes it performs, and the outputs it generates. In this study, the functional requirements outline the core features that AutoProject+ must provide to support online booking, vehicle customization, real-time tracking, payment verification, and administrative oversight for AutoProject-D Custom Garage.

User Authentication and Role Management. This module is essential for securing access and personalizing the experience for four distinct user roles: Customers, Administrators, Staff, and Mechanics. Customers use credentials to manage their bookings and vehicle profiles; Administrators oversee the entire operation; Staff handle walk-in encoding and support; and Mechanics access specific technical job orders. The system features secure login protocols and role-based access control (RBAC) to ensure data privacy.

Customer Dashboard. This serves as the primary interface for users to manage their automotive projects. It displays a real-time summary of active bookings, recent payment history, and a "Quick Track" status for vehicles currently in the garage. It also provides a direct link to the support ticketing system for any inquiries regarding ongoing customizations.

Customization Catalog. This module functions as a digital showcase where customers can browse various automotive services, such as body kits, custom paint, engine tuning, and interior upholstery. Each entry includes high-quality reference images, descriptions, and "starting at" price points. This allows customers to make informed choices before proceeding to the formal booking stage.

Booking and Estimation Engine. This is the core logic module where customers input their vehicle’s technical details (Make, Model, Year) and select specific customization packages. The system automatically calculates a Preliminary Cost Estimate based on stored service rates. This module ensures that both the customer and the garage have a clear, documented starting point for the project's scope and budget.

Administrative Management Portal. This module allows the garage owner or manager to review pending requests, evaluate garage floor capacity, and finalize cost estimates into formal quotes. It provides tools to assign specific mechanics to job orders and monitor the overall progress of all vehicles currently in the shop.

Technician Module (Mechanic Interface). Designed specifically for the technical team, this module displays assigned job orders with detailed work specifications. Mechanics use this interface to provide manual status updates (e.g., Queued, Ongoing, Paused, Completed) and add technical remarks or photos of the progress, which are then reflected on the customer's end.

Payment Integration. This module facilitates the secure payment of reservation fees. It integrates with external gateways like GCash and Maya, allowing customers to settle deposits digitally. The system automatically verifies the transaction via callback and updates the booking status to "Confirmed" once the payment is successful.

Real-Time Tracking System. This provides transparency by allowing customers to see exactly which stage their vehicle is in. From "Admin Review" and "Pending Payment" to "Work in Progress" and "Ready for Pickup," every status change triggered by the staff or mechanic is visible to the customer in real-time.

Hybrid Support (Ticketing System). Customers can submit support tickets for specific concerns or change requests. Staff members monitor this module to provide timely responses, bridging the gap between digital interaction and physical garage support. This ensures a high level of responsiveness and customer satisfaction.

Reports and Business Analytics. This module allows the Administrator to generate automated summaries of revenue, most requested services, mechanic productivity, and monthly booking volumes. These reports can be exported in PDF format, helping the business identify trends and optimize resource allocation.

Non-Functional Requirements

The following requirements are not directly concerned with the main function of the developed AutoProject+ web-based application but are concerned with the quality attributes needed by the product. The ISO 25010 standard is used by the researchers as the basis for the following requirements to ensure the system remains robust and efficient.

Functional Suitability. This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used in different conditions and circumstances. The design and features of the developed system, AutoProject+, focus on the digital transformation of garage operations, specifically online booking and real-time project tracking. The developed system enables users to inquire about services, book customization slots, and monitor the progress of their vehicles through a secure digital interface.

Performance Efficiency. This characteristic represents the performance of the developed system, which can perform efficiently in different conditions. The developed system requires a stable internet connection to calculate cost estimates in real-time and to ensure a smooth transition between the customer and administrator dashboards. The proponents ensure that the developed system can be used on any device that has a modern web browser. The system administrator can monitor overall garage capacity while mechanics can update job statuses without significant latency.

Compatibility. This characteristic represents the degree to which a product, system, or component can exchange information with other products, systems, or components and/or perform its required functions while sharing the same hardware or software environment. The user can book, inquire, and track their vehicle on any device with a web browser. Furthermore, the system is designed to exchange information accurately with external payment gateways like GCash and Maya to verify reservation fees.

Usability. This characteristic represents the effectiveness of the system, which the user can fully utilize with the goal of effectiveness, efficiency, and satisfaction in a specified context of use. The proponents ensure that the user can easily navigate the interface of the developed system, featuring a clean layout for the customization catalog and a straightforward dashboard. This ensures that even customers with minimal technical background can complete a booking successfully.

Reliability. This characteristic represents the reliability of the developed system because the component performs specific functions under specified conditions for a specified period of time. The proponents ensure that the developed system's automated logic—such as the cost estimation engine and scheduling checks—executes properly. The developed system features a notification service that informs both the customer and the mechanic of any changes in the booking or project status.

Maintainability. This characteristic represents the effectiveness and efficiency with which a product or system can be modified to improve it, correct it, or adapt it to changes in the environment and requirements. The developed system, AutoProject+, is built with a modular architecture so it can be fixed and maintained if bugs are detected. The administrator of AutoProject-D Custom Garage manages the core data, which includes editing the service catalog, updating material rates, and overseeing user accounts.

Security. This characteristic represents the degree to which a product or system protects information and data so that persons or other products or systems have the degree of data access appropriate to their types and levels of authorization. The collected data of the developed system, including customer profiles and vehicle details, remains confidential and is protected by role-based access control. Only authorized administrators can modify critical business settings or financial records.

Flexibility. This characteristic ensures that the system can adapt to changing requirements, such as new business processes, regulations, or technology updates. The proponents ensure that the data posted on the system, such as service pricing per customization, can be updated based on the current market rates of the garage. The developed system is designed to accommodate new service categories or additional payment methods with minimal changes to the underlying source code.

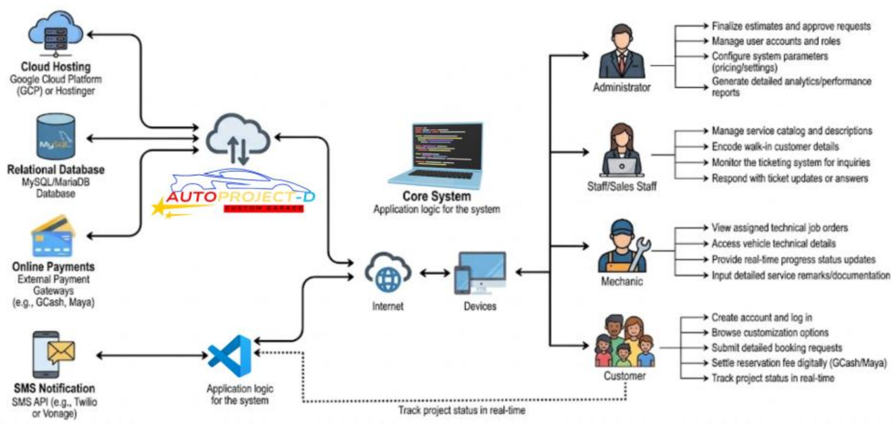
Safety. This characteristic focuses on safeguarding both users and the environment from potential harm caused by the system, primarily focusing on data safety and integrity. The proponents ensure that all gathered data from the user is handled through secure protocols to protect personal information. To prevent accidental data loss, the system incorporates confirmation prompts for critical actions and maintains regular database backups to ensure business continuity.

Requirements Specification

System Architecture

Figure 12

System Architecture



This figure illustrates the structural design of the AutoProject+ System, detailing the interconnectedness of core technologies, external service integrations, and the specific functional flows for each user role. The architecture is built as a centralized web-based platform where the Application Logic, developed using Visual Studio Code, acts as the primary coordinator between the cloud infrastructure and the end-user devices. Through internet connectivity, the system synchronizes data across the MySQL/MariaDB Relational Database, ensuring that vehicle specifications, booking statuses, and technical remarks are updated in real-time across all interfaces.

Customers access the system via web browsers on their personal devices to perform essential tasks such as account creation, browsing the customization catalog, and submitting detailed booking requests. The architecture facilitates a secure end-to-end transaction flow, allowing customers to settle reservation fees digitally through integrated Online Payment Gateways (GCash/Maya) and track their project's progress in real-time.

Mechanics utilize the system to manage the technical execution of customization projects. Their interface provides direct access to assigned job orders and vehicle technical details. As the work progresses, mechanics input real-time status updates and detailed service remarks, which are immediately processed by the core system logic and stored in the relational database for customer visibility and administrative documentation.

Staff and Sales Personnel serve as the bridge for walk-in operations and customer inquiries. They use authenticated web accounts to manage the service catalog, encode details for customers who visit the garage in person, and monitor the integrated ticketing system. This role ensures high responsiveness by providing timely updates or answers to customer support tickets, all managed through the central application server.

Administrators maintain overarching control of the environment through elevated access privileges. Their responsibilities include the finalization of cost estimates, approval of service requests, and management of user accounts and roles. The architecture provides the Administrator with comprehensive analytics and performance reports, drawing data from the database to visualize business trends and garage efficiency.

This architecture enables a cohesive operational environment by integrating Cloud Hosting (GCP/Hostinger) and SMS Notification APIs (Twilio/Vonage) with the core system logic. The use of a robust relational database and role-based access control ensures that AutoProject-D Custom Garage maintains data integrity and operational transparency, providing a seamless experience from the initial digital inquiry to the final vehicle release.

Hardware and Software Specification

Commented [2]: Revise this if you are going to start the development.

Table 6

The Software and Hardware Specification for the development of AutoProject+.

Software Specification (Development)	Name	Minimum	Recommended
Operating System	Windows	Windows 10 (64-bit)	Windows 11 (64-bit)
Wireframing	UI/UX Design	Figma (Web-based)	Figma (Latest version)
Programming Language	Core Languages	PHP 7.4, JavaScript ES6	PHP 8.2+, JavaScript ES2022
Frontend	Web Interface	HTML5, CSS3, Bootstrap 5	HTML5, CSS3, Tailwind CSS
Backend	Server Logic	Native PHP / Laravel 9.x	Laravel 10.x Framework
Infrastructure	Local/Cloud	XAMPP (Apache)	Google Cloud Platform / Hostinger
IDE	Development Tool	Visual Studio Code v1.70+	Visual Studio Code (Latest)
Database	Data Store	MySQL 5.7	MySQL 8.0 / MariaDB
API	Notifications	Twilio/Vonage SMS API	Twilio/Vonage (REST Integration)
Hardware Specification (Development)	Component	Minimum	Recommended
Processor (Cores)	CPU	Intel Core i5 (4 Cores)	Intel Core i7 / Ryzen 7 (8 Cores)
Device RAM	Memory	8 GB DDR4	16 GB DDR4/DDR5
Storage	Disk Space	256 GB HDD	512 GB SSD (NVMe)

Table 6 outlines the system requirements for developing the AutoProject+ web-based system. It specifies both minimum and recommended configurations for the development environment. Windows 11 is recommended for a stable development experience using Visual Studio Code as the primary IDE. The backend is powered by PHP, while the frontend utilizes HTML5, CSS3, and JavaScript for a responsive user interface. MySQL serves as the database for storing user, booking, and vehicle records, while Figma was used for initial wireframing to align with the system's conceptual design.

Table 7

Hardware Specification Needed for Implementing the System (End User Devices)

Hardware Specification (Web Access – Windows, macOS, Linux)	Name	Minimum	Recommended
Processor	CPU Speed	Dual-core 2.0GHz	Quad-core 2.5GHz+
Device RAM	Memory	4 GB	8 GB or higher
Network	Connectivity	5 Mbps (Stable)	20 Mbps+ Broadband
Display	Resolution	1280×720 (HD)	1920×1080 (Full HD)
Hardware Specification (Mobile – for Real-Time Tracking)	**Name**	**Minimum**	**Recommended**
Processor	Mobile CPU	Quad-core 1.8GHz	Octa-core 2.2GHz+
Device RAM	Memory	3 GB LPDDR4	6 GB or higher
Network	Mobile Data	4G LTE / Wi-Fi	5G / Dual-band Wi-Fi
Display	Screen	5.5" Touchscreen	6.5" AMOLED/IPS Display

Table 7 outlines the hardware requirements for implementing AutoProject+ on end-user devices, covering desktops for administrative tasks and mobile devices for customer tracking. For web users (Staff and Admin), a quad-core processor and 8GB RAM are recommended to handle the high volume of data in the dashboard efficiently. For mobile users (Customers and Mechanics), a minimum of 3GB RAM and a stable 4G connection are required to ensure that real-time status updates and notifications are received without delay.

Table 8

Software Specification Needed for Implementing the System (End User Devices)

Software Specification (Web Access – Windows, macOS)	**Name**	**Minimum**	**Recommended**
Operating System	Desktop OS	Windows 10, macOS 11	Windows 11, macOS 13+
Web Browser	Interface	Chrome 95+, Edge 95+	Chrome (Latest), Safari 16+
Bandwidth	Internet	5 Mbps	15 Mbps for faster report loading
Software Specification (Mobile Access – Android, iOS)	Platform	Minimum	Recommended
Android (OS)	Mobile OS	Android 9.0 (Pie)	Android 12.0 or higher
Apple (iOS)	Mobile OS	iOS 14.0	iOS 16.0 or higher

Table 8 defines the software requirements for end-users to interact with the AutoProject+ system. It specifies that modern web browsers like Chrome and Edge are required for accessing the booking and management modules. For mobile users, the system supports older versions of Android and iOS to maintain accessibility, though the latest versions are recommended for optimal security and notification performance. This ensures that the system remains accessible across a wide range of devices while maintaining high standards for data safety.

Test Case Document (TCD)

The test case document showed the paradigm that was used in testing Autoproject+: A Web-Based Online Service Booking And Cost Estimation System With Hybrid Customer Support For Autoproject-d Custom Garage.

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Table 9

Test Case Document for AutoProject+ (Authentication Module)

TEST CASE AUTOPROJECT+: A WEB-BASED ONLINE SERVICE BOOKING AND COST ESTIMATION SYSTEM WITH HYBRID CUSTOMER SUPPORT FOR AUTOPROJECT-D CUSTOM GARAGE				
Test Case #	**001**	**Test Case Name**	**System Module 1**	**Page 1 of 12**
System	AutoProject+	**Subsystem**	Authentication Module	
Designed by	**Charise Anne Lacdao**	**Designed Date:**		
Executed by		**Execution Date:**		
Short Description	Verification of the login portal, credential validation, and session security.			
Pre-condition: 1. The user has an active account (Admin, Staff, Mechanic, or Customer). 2. The user is currently at the landing page and is not yet authenticated. 3. The database server is online to verify credentials.				
Step	**Action**	**Expected System Output**	**Pass/Fail**	**Remarks**
1	Enter URL in browser address bar.	The system loads the landing page with the "Login" button..		
2	Click the "Login" button.	The system redirects to the Login Form page.		
3	Enter a valid Admin username and password.	System validates credentials and loads Admin Dashboard.		
4	Enter an incorrect password.	System displays error: "Incorrect password."		
5	Click "Logout" from the sidebar.	System ends session and redirects to login page.		
Post condition: 1. The user is successfully redirected to their specific role-based dashboard. 2. An active session token is created for the user.				

Table 10

Test Case Document for AutoProject+ (Admin Dashboard Module)

TEST CASE AUTOPROJECT+: A WEB-BASED ONLINE SERVICE BOOKING AND COST ESTIMATION SYSTEM WITH HYBRID CUSTOMER SUPPORT FOR AUTOPROJECT-D CUSTOM GARAGE				
Test Case #	**002**	**Test Case Name**	**System Module 2**	**Page 2 of 12**
System	AutoProject+	**Subsystem**	Admin Dashboard Module	
Designed by	**Charise Anne Lacdao**	**Designed Date:**		
Executed by		**Execution Date:**		
Short Description	Testing the accuracy of real-time data visualization on the main admin landing page.			
Pre-condition: 1. The user is logged in with "Administrator" privileges. 2. Existing data (bookings/revenue) exists in the database to populate charts.				
Step	**Action**	**Expected System Output**	**Pass/Fail**	**Remarks**
1	Access the Dashboard page.	Dashboard loads with summary cards and charts.		
2	Observe "Total Bookings" counter.	Displays correct count of records from the database.		
3	Hover over the Revenue Line Graph.	Tooltips appear showing specific monthly earnings.		
4	Click "Refresh" on analytics.	The charts update without reloading the entire page.		
Post condition: 1. The Administrator has reviewed the current business metrics accurately. 2. No data manipulation occurred during the viewing process.				

Table 11

Test Case Document for AutoProject+ (Service Catalog Module)

TEST CASE AUTOPROJECT+: A WEB-BASED ONLINE SERVICE BOOKING AND COST ESTIMATION SYSTEM WITH HYBRID CUSTOMER SUPPORT FOR AUTOPROJECT-D CUSTOM GARAGE				
Test Case #	**003**	**Test Case Name**	**System Module 3**	**Page 3 of 12**
System	AutoProject+	**Subsystem**	Service Catalog Module	
Designed by	**Charise Anne Lacdao**	**Designed Date:**		
Executed by		**Execution Date:**		
Short Description	Testing the addition, editing, and deletion of garage customization services.			
Pre-condition: 1. The user is logged in as an Administrator or authorized Staff. 2. The user is on the "Manage Services" page.				
Step	**Action**	**Expected System Output**	**Pass/Fail**	**Remarks**
1	Click "Add New Service".	A form appears for Service Name, Description, and Price.		
2	Input "Full Body Ceramic" and price.	The new service is added to the public catalog.		
3	Click "Edit" on an existing service.	The price or description fields become editable.		
4	Click "Delete" on a discontinued service.	System asks for confirmation and removes the item.		
Post condition: 1. The public customization catalog reflects the updated service list. 2. Booking estimates will now use the updated pricing data.				

Table 12

Test Case Document for AutoProject+ (Customer Booking Module)

TEST CASE AUTOPROJECT+: A WEB-BASED ONLINE SERVICE BOOKING AND COST ESTIMATION SYSTEM WITH HYBRID CUSTOMER SUPPORT FOR AUTOPROJECT-D CUSTOM GARAGE				
Test Case #	**004**	**Test Case Name**	**System Module 4**	**Page 4 of 12**
System	AutoProject+	**Subsystem**	Customer Booking Module	
Designed by	**Charise Anne Lacdao**	**Designed Date:**		
Executed by		**Execution Date:**		
Short Description	Testing the flow from vehicle detail entry to preliminary cost estimation.			
Pre-condition: 1. The user is logged in as a "Customer". 2. The customer has a vehicle ready for registration in the system.				
Step	**Action**	**Expected System Output**	**Pass/Fail**	**Remarks**
1	Click "Book Service".	System loads the multi-step booking form.		
2	Input Vehicle Make/Model/Year.	Data is temporarily saved for the current session.		
3	Select "Custom Paint" service.	System updates the running total estimate.		
4	Click "Confirm & Submit Request".	Booking is saved as "Pending" and ID is generated.		
Post condition: 1. A new booking record is created in the database. 2. An automated notification is triggered for the Admin/Staff.				

Table 13

Test Case Document for AutoProject+ (Mechanic Task Module)

TEST CASE AUTOPROJECT+: A WEB-BASED ONLINE SERVICE BOOKING AND COST ESTIMATION SYSTEM WITH HYBRID CUSTOMER SUPPORT FOR AUTOPROJECT-D CUSTOM GARAGE				
Test Case #	**005**	**Test Case Name**	**System Module 5**	**Page 5 of 12**
System	AutoProject+	**Subsystem**	Mechanic Task Module	
Designed by	**Charise Anne Lacdao**	**Designed Date:**		
Executed by		**Execution Date:**		
Short Description	Testing status transitions and technical logging for assigned jobs.			
Pre-condition: 1. The user is logged in as a "Mechanic". 2. At least one Job Order has been assigned to this specific mechanic.				
Step	**Action**	**Expected System Output**	**Pass/Fail**	**Remarks**
1	Open "My Tasks" page.	A list of assigned vehicles is displayed.		
2	Change status to "Ongoing".	Status changes and a timestamp is recorded.		
3	Input technical remarks.	Remarks are saved and appended to the service record.		
4	Set status to "Completed".	System closes the job and notifies the customer.		
Post condition: 1. The Job Order is marked as finished in the central database. 2. The vehicle is moved to the "Ready for Release" queue.				

Table 14

Test Case Document for AutoProject+ (Payment Verification Module)

TEST CASE AUTOPROJECT+: A WEB-BASED ONLINE SERVICE BOOKING AND COST ESTIMATION SYSTEM WITH HYBRID CUSTOMER SUPPORT FOR AUTOPROJECT-D CUSTOM GARAGE				
Test Case #	**006**	**Test Case Name**	**System Module 6**	**Page 6 of 12**
System	AutoProject+	**Subsystem**	Payment	

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			Verification Module	
Designed by	**Charise Anne Lacdao**	**Designed Date:**		
Executed by		**Execution Date:**		
Short Description	Verification of payment gateway integration, reference number entry, and status confirmation.			
Pre-condition: 1. The user (Customer) has an "Approved" booking status. 2. The user has access to a digital wallet (GCash/Maya). 3. The system is connected to the Payment Gateway API or manual verification logic.				
Step	**Action**	**Expected System Output**	**Pass/Fail**	**Remarks**
1	Click "Pay Reservation Fee" button.	System displays the Payment Modal with QR code/options.		
2	Input a valid 13-digit Reference Number.	System accepts input and proceeds to verification.		
3	Click "Confirm Payment".	System validates the transaction and updates status to "Paid".		
4	Input a previously used Reference Number.	System displays error: "Reference number already used."		
5	View "Payment History" page.	The recent transaction appears with Date, Amount, and ID.		
Post condition: 1. The booking status is automatically updated to "Confirmed" in the database. 2. A digital receipt is generated and available for download by the customer.				

Table 15

Test Case Document for AutoProject+ (Support Ticketing Module)

TEST CASE AUTOPROJECT+: A WEB-BASED ONLINE SERVICE BOOKING AND COST ESTIMATION SYSTEM WITH HYBRID CUSTOMER SUPPORT FOR AUTOPROJECT-D CUSTOM GARAGE				
Test Case #	**007**	**Test Case Name**	**System Module 7**	**Page 7 of 12**
System	AutoProject+	**Subsystem**	Support Ticketing	

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			Module	
Designed by	**Charise Anne Lacdao**	**Designed Date:**		
Executed by		**Execution Date:**		
Short Description	Testing the creation, response, and resolution of support tickets between Customer and Staff.			
Pre-condition: 1. The Customer is logged in and has an active or past booking. 2. Staff is logged in to monitor the support dashboard.				
Step	**Action**	**Expected System Output**	**Pass/Fail**	**Remarks**
1	Navigate to "Support" page and click "Create Ticket".	A form appears with Subject, Category, and Message fields.		
2	Submit a ticket regarding "Paint Warranty".	Ticket is saved and assigned a unique Ticket ID.		
3	Staff clicks "Open Ticket" from their dashboard..	Staff can view the customer's full message and booking info.		
4	Staff types a reply and clicks "Send"	The reply is visible on the Customer's support interface.		
5	Staff clicks "Mark as Resolved".	Ticket status changes to "Closed" and is moved to archives.		
Post condition: 1. A complete communication log is stored for the specific project. 2. The customer receives an email/SMS notification for the staff's reply.				

Table 16

Test Case Document for AutoProject+ (Reports & Analytics Module)

TEST CASE AUTOPROJECT+: A WEB-BASED ONLINE SERVICE BOOKING AND COST ESTIMATION SYSTEM WITH HYBRID CUSTOMER SUPPORT FOR AUTOPROJECT-D CUSTOM GARAGE				
Test Case #	**008**	**Test Case Name**	**System Module 8**	**Page 8 of 12**
System	AutoProject+	**Subsystem**	Reports & Analytics Module	
Designed by	**Charise Anne**	**Designed Date:**		

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	Lacdao			
Executed by	**Sheila R. Folgar**	**Execution Date:**		
Short Description	Testing the generation and export of business performance reports.			
Pre-condition: 1. The user is logged in as an "Administrator". 2. The database contains at least one month's worth of transaction data.				
Step	**Action**	**Expected System Output**	**Pass/Fail**	**Remarks**
1	Navigate to the "Reports" page.	System displays filter options (Date Range, Service Type).		
2	Select "Monthly Revenue" and click "Generate".	A table/chart appears summarizing earnings for the month.		
3	Click "Export to PDF".	A PDF file is generated containing the professional report layout.		
4	Select "Mechanic Performance" report.	The reply is visible on the Customer's support interface.		
Post condition: 1. Data accuracy is verified against raw database records. 2. The generated files are formatted correctly for physical printing/filing.				

Table 17

Test Case Document for AutoProject+ (System Logs & Audit Trail)

TEST CASE AUTOPROJECT+: A WEB-BASED ONLINE SERVICE BOOKING AND COST ESTIMATION SYSTEM WITH HYBRID CUSTOMER SUPPORT FOR AUTOPROJECT-D CUSTOM GARAGE				
Test Case #	**009**	**Test Case Name**	**System Module 9**	**Page 9 of 12**
System	AutoProject+	**Subsystem**	System Logs & Audit Trail	
Designed by	**Charise Anne**	**Designed Date:**		

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	Lacdao			
Executed by	**Sheila R. Folgar**	**Execution Date:**		
Short Description	Verification that all critical system actions are logged for administrative review.			
Pre-condition: 1. The user is logged in as "Administrator". 2. Several actions (logins, deletions, updates) have been performed by other users.				
Step	**Action**	**Expected System Output**	**Pass/Fail**	**Remarks**
1	Navigate to "System Logs" page.	A chronological list of all system activities is displayed.		
2	Search for a specific "User ID" in the logs.	All actions performed by that user are filtered and shown.		
3	Check for "Failed Login" entries.	System correctly logs the timestamp and IP of failed attempts.		
4	Verify "Data Deletion" logs.	System shows who deleted a record and when.		
Post condition: 1. The Administrator has a complete, uneditable trail of system usage. 2. High-level security accountability is maintained.				

Table 18

Test Case Document for AutoProject+ (Vehicle Registration Module)

TEST CASE **AUTOPROJECT+: A WEB-BASED ONLINE SERVICE BOOKING AND COST ESTIMATION SYSTEM WITH HYBRID CUSTOMER SUPPORT FOR AUTOPROJECT-D CUSTOM GARAGE**				
Test Case #	**010**	**Test Case Name**	**System Module 10**	**Page 10 of 12**
System	AutoProject+	**Subsystem**	Vehicle Registration Module	
Designed by	**Charise Anne**	**Designed Date:**		

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	Lacdao			
Executed by		**Execution Date:**		
Short Description	Verification of adding, viewing, and updating vehicle details for customization profiles.			
Pre-condition: 1. The user is logged in as a "Customer". 2. The user has valid vehicle identification documents (OR/CR) for data entry. 3. The "My Vehicles" page is accessible via the customer sidebar.				
Step	**Action**	**Expected System Output**	**Pass/Fail**	**Remarks**
1	Click "Add New Vehicle".	System displays a form for Plate #, Model, Year, and Color.		
2	Input a duplicate Plate Number.	System displays error: "Plate number already registered."		
3	Input valid vehicle details and click "Save".	The vehicle is added to the customer's profile gallery.		
4	Click "Edit" on an existing vehicle entry.	Fields become editable to update details like "Color".		
5	Click "Delete" on a vehicle record.	System prompts for confirmation before removal.		
Post condition: 1. The customer's vehicle list is updated in the database. 2. These vehicles are now selectable in the Booking Module dropdown menu.				

Table 19

Test Case Document for AutoProject+ (Notification Settings Module)

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TEST CASE AUTOPROJECT+: A WEB-BASED ONLINE SERVICE BOOKING AND COST ESTIMATION SYSTEM WITH HYBRID CUSTOMER SUPPORT FOR AUTOPROJECT-D CUSTOM GARAGE				
Test Case #	**011**	**Test Case Name**	**System Module 11**	**Page 11 of 12**
System	AutoProject+	**Subsystem**	Notification Settings Module	
Designed by	**Charise Anne**	**Designed Date:**		

	Lacdao			
Executed by		**Execution Date:**		
Short Description	Testing the toggling of system alerts and verification of SMS/Email triggers.			
Pre-condition: 1. The user is logged in (any role). 2. The SMS API (Twilio/Vonage) is configured and has active credits. 3. The user has a valid mobile number saved in their profile.				
Step	**Action**	**Expected System Output**	**Pass/Fail**	**Remarks**
1	Navigate to "Account Settings > Notifications".	System shows toggles for SMS, Email, and Push alerts.		
2	Toggle "SMS Alerts for Status Updates" to ON.	Preference is saved; a confirmation "Settings Updated" appears.		
3	Trigger a status change (e.g., Job Completed).	The user receives an automated SMS on their registered device.		
4	Toggle "Email Notifications" to OFF.	System stops sending emails for non-critical alerts.		
Post condition: 1. User notification preferences are updated in the User table. 2. The automated communication logs reflect successful alert delivery.				

Table 20

Test Case Document for AutoProject+ (User Profile Management Module)

TEST CASE AUTOPROJECT+: A WEB-BASED ONLINE SERVICE BOOKING AND COST ESTIMATION SYSTEM WITH HYBRID CUSTOMER SUPPORT FOR AUTOPROJECT-D CUSTOM GARAGE				
Test Case #	**012**	**Test Case Name**	**System Module 12**	**Page 12 of 12**
System	AutoProject+	**Subsystem**	User Profile Management	

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			Module	
Designed by	**Charise Anne Lacdao**	**Designed Date:**		
Executed by		**Execution Date:**		
Short Description	Testing the update of personal information and secure password changes.			
Pre-condition: 1. The user is logged in to their respective account. 2. The user is on the "My Profile" page.				
Step	**Action**	**Expected System Output**	**Pass/Fail**	**Remarks**
1	Upload a new "Profile Picture".	The image is uploaded, resized, and displayed instantly.		
2	Update "Contact Number" and click "Save".	The database reflects the new number; success toast appears.		
3	Navigate to "Security > Change Password".	System prompts for Current, New, and Confirm Password.		
4	Input a New Password that is too short.	System displays: "Password must be at least 8 characters."		
5	Input valid Current and New passwords.	Password is encrypted and updated; user remains logged in.		
Post condition: 1. The user's personal identification data is accurately maintained. 2. Account security is reinforced through updated credentials.				

Evaluation Plan

The developed AutoProject+: A Web-Based Custom Garage Management System for AutoProject-D was evaluated using the ISO 25010 quality model, which provides a comprehensive framework for assessing software quality. This standard encompasses nine key characteristics: functional suitability, performance efficiency, compatibility, usability,

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reliability, security, maintainability, flexibility, and safety. These dimensions served as the technical basis for evaluating the system's ability to meet the specialized requirements of automotive customization and administrative oversight effectively.

To gather empirical data, respondents were selected from two primary stakeholder groups: the garage management/staff (including mechanics) and regular clients of AutoProject-D. The proponents conducted structured interviews and system walkthroughs to identify existing challenges in the manual project tracking process and to gather insights into potential areas for digital improvement. The qualitative and quantitative data gathered were systematically analyzed to inform the final system refinements and ensure alignment with user expectations and ISO 25010 criteria.

A purposive sampling technique was employed, targeting individuals who had direct and relevant experience with the garage's services or automotive management. This non-probability sampling method was selected to ensure that the feedback gathered would be meaningful and based on actual system use-cases. A total of 20 respondents were chosen, as their specific roles and experience in the automotive industry made them suitable to provide informed insights for the study.

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Table 21

Scales for the Evaluation Ratings

Weight	Interval	Verbal Interpretation

4	3.25-4.00	Very Acceptable
3	2.50-3.24	Highly Acceptable
2	1.75-2.49	Acceptable
1	1.00-1.74	Not Acceptable

The weighted mean of each item in the user evaluation of the developed system will be calculated using the formula:

$$\mathbf{WM} = \frac{\mathbf{4(f) + 3(f) + 2(f) + 1(f)}}{\mathbf{N}}$$

Where:

WM = Weighted Mean

f = Frequency of Response per weight

N = Total Number of Respondents

Integration Testing. This testing technique is best used for garage management systems to verify the interaction between the end-user (Customer/Mechanic) and the backend administrative logic. This technique enabled the testing of the system's features, such as the transition from a booking request to an active job order, as a group to determine whether they worked together effectively under various operational scenarios.

The goal of integration testing for AutoProject+ was to determine whether the modules, such as the Cost Estimation Engine and the Payment Verification API, performed

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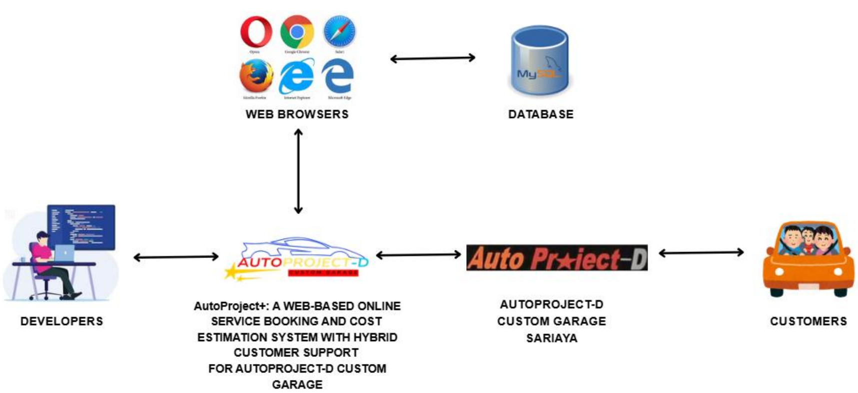
as expected when integrated together. It checked if data flowed properly between the customer's selection and the mechanic's task list, ensuring interfaces were compatible and dependencies were resolved without errors. Integration testing revealed defects such as communication delays between components or improper data exchange regarding vehicle specifications, allowing the proponents to optimize the system for the garage's high-pressure environment.

Deployment Plan

Figure 13

AutoProject+ Deployment Plan



This figure illustrates the overall architecture and interaction flow of the AutoProject+ Web-Based Online Service Booking and Cost Estimation System designed for AutoProject-D Custom Garage in Sariaya. It presents how the major components of the system work together to support online booking, customer interaction, and service management. The diagram shows the relationship between developers, the web-based application, the database, the physical garage operations, and the customers who use the system.

On the left side of the diagram are the developers, who are responsible for designing, developing, and maintaining the AutoProject+ system. Developers create and update the system's code, implement new features, and ensure that the platform functions properly. Once the system is developed, it is deployed so that it can be accessed through different web browsers such as Google Chrome, Opera, Mozilla Firefox, Microsoft Edge, and Safari. These browsers serve as the primary interface where users interact with the web-based platform.

At the top right of the diagram is the database, represented by MySQL. The database is responsible for storing all system information, including customer accounts, booking details, vehicle information, service requests, and transaction records. The web browser communicates with the database through the AutoProject+ system to retrieve and store data whenever users perform actions such as creating bookings, viewing service details, or updating records.

At the center of the architecture is the AutoProject+ web-based system, which acts as the core platform connecting all components. This system processes user requests, manages bookings, calculates cost estimates, and handles communication between the user interface and the database. It functions as the main application that supports hybrid customer service by combining online digital services with the physical operations of the garage.

On the right side of the diagram is AutoProject-D Custom Garage in Sariaya, which represents the physical location where vehicle customization and repair services are performed. The garage receives service requests and job orders that are submitted through the AutoProject+ system. Mechanics and staff use the information from the system to perform the required vehicle services and update the service status.

Customers interact with the system by accessing the platform through their web browsers. They can browse available services, submit booking requests, receive cost estimates, and communicate with the garage. Customers are connected to the AutoProject-D system through the web platform, allowing them to conveniently manage their service appointments without visiting the garage immediately.

The figure demonstrates how AutoProject+ integrates developers, web browsers, the database, the garage operations, and customers into a unified system. This architecture enables efficient service booking, accurate cost estimation, and improved communication between the garage and its clients.

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APPENDICES

APPENDIX A

Letter for Request for Interview

January 17, 2025

Makol E. Pasumbal
Owner
AutoProject-D Custom Garage
Poblcaion 3, Sariaya, Quezon

Dear Sir:

Warmest greetings!

The undersigned is a third-year student taking up a Bachelor of Science in Information Technology at CSTC-College of Sciences, Technology, and Communication Inc. Sariaya Campus, and I am presently working on their course subject, Capstone Project & Research 1 (CAPS 301).

In this regard, we would like to request that you be one of the experts who will be interviewed to collect your comments and suggestions that would greatly contribute to the success of our proposed capstone project. Rest assured that your responses will be kept confidential and will only be used for educational purposes.

We highly appreciate any assistance and support you can provide in our research endeavor. Thank you for sharing your extra time and expertise.

Best regards,

Charise Anne Lacdao
09518644486

Karl Angelo Venua
09481270864

Ryu Jed Vi[illegible]
0907[illegible]5905
Proponents

Noted:

JORGE [illegible]ADER, MIT
Dean, School of Information Technology
CSTC Inc., Sariaya Campus

Received:

APPENDIX B

Collaboration Letter

COLLABORATION AGREEMENT

This Collaboration Agreement is entered into on March 27, 2026, between **Charise Anne D. Lacdao, Karl Angelo G. Venua** and **Ryu Jed M. Virtucio** hereinafter referred to as the "Researchers." and **Makol E. Pasumbal**, hereinafter referred to as the "Client".

The purpose of this Agreement is to outline the terms and conditions governing the collaboration between the Client and the Developer for the development of an IT Capstone Research Project, further detailed below:

1. **Project Description**: The project involves the design and development of AutoProject+: A Web-Based Online Service Booking and Cost Estimation System with Hybrid Customer Support for AutoProject-D Custom Garage. The system is intended to provide a centralized platform for managing both online reservations and walk-in customers through digital scheduling and service management. It includes an automated cost estimation feature that generates preliminary service quotations based on vehicle and service details. In addition, the system supports customer interaction through ticket-based communication and service tracking, while implementing role-based access control for administrators, staff, and mechanics. It also provides administrative reporting features for monitoring bookings, service activities, cost estimations, and estimated sales and revenue performance.

2. **Roles and Responsibilities**:

   a. **Client Responsibilities**: The Client agrees to provide necessary project requirements, resources, and feedback in a timely manner.

   b. **Developer Responsibilities**: The Developer agrees to develop the software according to the specifications provided by the Client and to provide regular updates on the progress of the project.

3. **Timeline**: The project timeline will be as follows: **Academic Year 2026 – 2027**

4. **Intellectual Property Rights**: All intellectual property rights, including but not limited to software code, design, and documentation, developed during the project shall belong to the Researchers and deployment will be optionally endorsed to the Client upon completion of the project.

5. **Confidentiality**: Both parties agree to keep all project-related information confidential and not to disclose any confidential information to third parties without prior written consent.

6. **Termination**: Either party may terminate this Agreement with written notice to the other party if there is a material breach of any provision of this Agreement.

7. **Governing Law**: This Agreement shall be governed by and construed in accordance with the R.A. 10173 Data Privacy Act of 2012 and R.A. 8293 Intellectual Property Law.

8. **Amendments**: Any amendments to this Agreement must be made in writing and signed by both parties.

9. **Entire Agreement**: This Agreement constitutes the entire understanding between the Client and the Researchers and supersedes all prior agreements, whether written or oral, relating to the subject matter herein.

IN WITNESS WHEREOF, the parties hereto have executed this Agreement as of the date first above written.

Charise Anne D. Lacdao

Karl Angelo G. Venua

Ryu Jed M. Virtucio

Researchers

Noted:

SHIELA R. FOLGAR
Capstone Adviser

MAKOL E. PASUMBAL
Client

Approved:

JORGE T. JADER, MIT
Dean, School of Information Technology
CSTC- Sariaya, Quezon

APPENDIX C

CBA Request Letter

February 4, 2026

Mark Anthony C. Delgado, CPA, MBA
Program Head – Accountancy
CSTC Inc., Sariaya Campus
Brgy. Poblacion 3, Arellano St. Sariaya, Quezon

Dear Sir,

We, the 3 rd year students of Bachelor of Sciences in Information Technology at CSTC College of Sciences, Technology and Communications Inc., Sariaya Campus are currently working on our proposal entitled, **AutoProject+: A Web-Based Online Service Booking and Cost Estimation System with Hybrid Customer Support for AutoProject-D Custom Garage.** as part of our academic curriculum.

Your expertise in analytics could greatly assist us in developing our proposed application. Additionally, we are interested in understanding how Material Requirements Planning (MRP) improves the proposed system feasibility.

We kindly request your availability for a Cost-Benefit Analysis validation and consultation. We are flexible and willing to accommodate your schedule at a time and place that is convenient for you. Should you have any questions or require further information, please do contact us. Your time and consideration are greatly appreciated.

Sincerely,

Lacdao, Charise Anne D.
Venua, Karl Angelo
Virtucio, Ryu Jed

Researchers

Noted:

JORGE T. JADER, MIT
Capstone Research Adviser
School of Information Technology

Received:

Mark Anthony C. Delgado

APPENDIX D

Cost Benefit Analysis (CBA)

Cost - Benefits Analysis

	0	1	2	3	4	5	Total
Total Benefits	0	₱500,000	₱500,000	₱500,000	₱500,000	₱500,000	₱2,500,000
Discount Rate (10%)	1	0.909090909	0.826446281	0.751314801	0.683013455	0.620921323	
NPV of Benefits	0	₱454,545	₱413,223	₱375,657	₱341,507	₱310,461	₱1,895,393
Total NPV of All Benefi	0	454,545.45	867,768.60	1,243,426.00	1,584,932.72	1,895,393.38	**₱1,895,393.38**
One-Time Cost	**₱149,197**						
Total Recurring Cost	0	₱19,220	₱19,220	₱19,220	₱19,220	₱19,220	₱96,100
Discount Rate (10%)	1	0.909090909	0.826446281	0.751314801	0.683013455	0.620921323	
NPV of Recurring Cost	0	₱17,472.73	₱15,884	₱14,440	₱13,128	₱11,934	₱72,859
Total NPV of All Cost	₱149,197	₱166,670	₱182,554	₱196,994	₱210,122	₱222,056	**₱222,056**
Net Income		437,072.73	₱397,339	₱361,217	₱328,379	₱298,527	

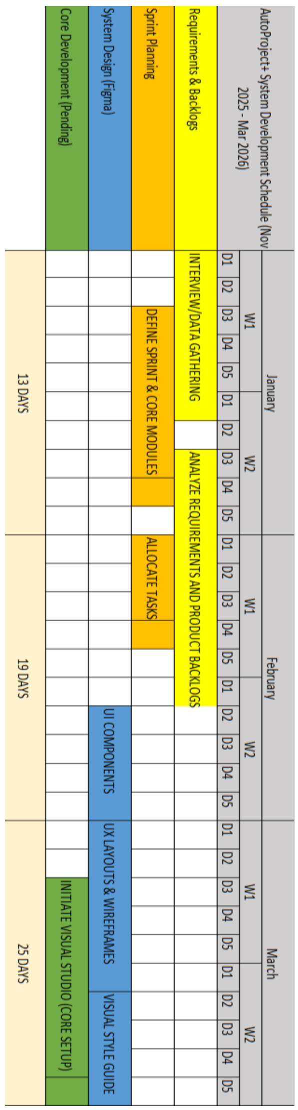
Overall NPV of All Benefits and Cost	**₱1,673,337**
Return of Investment (Net Income/Investment)	**293%**

Payback Period

Year	Net Income	
1	₱437,072.73	0.34
2	₱397,338.84	
		0.34 years

APPENDIX E

Gantt Chart



Commented [8]: This is INCORRECT!

Provide appropriate software methodology that align to your Gantt chart.

MAJOR REVISION!

APPENDIX G

User’s Manual Guide

APPENDIX F

NARRATIVE REPORT

The researchers conducted a series of semi-structured interviews with Makol Pasumbal, the owner and manager of AutoProject-D Custom Garage, on January 17, 2026, to gather essential information for the development of the proposed AutoProject+ system. The purpose of these interviews was to understand the background of the business, analyze its current operational workflow, and identify key challenges that could be addressed through a digital solution. Through these discussions, the researchers were able to gain a deeper understanding of the garage's daily operations and the issues affecting its efficiency. During the interview, Mr. Pasumbal shared the history and origin of the business, highlighting its strong connection to Project-D, a well-known automotive customization group from Japan. He explained that he was granted exclusive permission to establish the brand in the Philippines, making the Sariaya branch the only authentic and recognized branch in the country. This distinction has played a significant role in building the shop's reputation and attracting automotive enthusiasts from different regions, contributing to its steady growth and popularity. The organizational structure of the garage was also discussed, revealing a well-defined team composed of management, technical staff, and office personnel. Mr. Pasumbal oversees the entire operation, ensuring that both administrative and technical processes run smoothly. The technical team consists of skilled mechanics specializing in body works, paint application, and engine repair, while the office staff handle customer inquiries and manage transactions manually. Despite having a competent team, the current workflow relies heavily on traditional methods, which limits overall efficiency. Several operational challenges were identified during the interview. The increasing demand for services has resulted in a high volume of projects that often exceed the garage's capacity. Although the business focuses on customization, it also accommodates minor maintenance services, which adds to the workload without a proper system to filter or prioritize requests. Another major concern is the lack of a designated parking area, leading to vehicles being parked along streets and sidewalks, causing congestion and inconvenience within the surrounding area. Additionally, the absence of a structured scheduling system forces the business to rely on informal communication such as messaging and verbal agreements, resulting in overlapping appointments and disorganized workflows. To address these issues, the researchers proposed the development of AutoProject+, a web-based booking and cost estimation system designed to improve operational efficiency. The system introduces a digital booking feature that allows customers to schedule appointments in advance, helping to regulate the number of vehicles present at any given time and reduce overcrowding. It also includes an automated cost estimation tool that provides consistent and transparent pricing based on specific service requirements, minimizing misunderstandings between the business and its clients. Furthermore, a ticket-based support system was suggested to streamline communication and reduce unnecessary physical visits for simple concerns. The system also incorporates role-based access to ensure that the owner, staff, and mechanics can efficiently monitor and manage job orders. After presenting the proposed solution, Mr. Pasumbal shared constructive feedback and suggested additional features that could further enhance the system. His insights emphasized the importance of improving digital infrastructure and maintaining organized service records to support long-term growth. The researchers considered these recommendations and integrated them into the system design, ensuring that AutoProject+ will serve as a practical and sustainable solution for modernizing the operations of the garage.

Commented [9]: Kapag ng data gathering ang researchers hindi naka pambahay.

This is inappropriate.